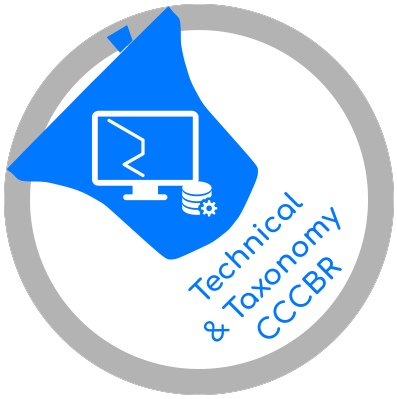


The Central Council of Church Bell Ringers

Framework for Method Ringing

Edition 2

January 26, 2022

Contents

	Page
1. Introduction	**1**
A. Background	1
B. Material	1
C. Overview of the Framework	1
D. Framework Principles	2
2. Versioning	**3**
3. Fundamentals of Method Ringing	**4**
A. Stages	4
B. Rows	4
C. Changes	6
D. Blocks	7
E. Methods	9
F. Calls	11
G. Compositions	12
H. Cover Bells	12
I. Touches	14
J. Truth	15
K. Performances	17
4. Method Classification	**18**
A. Definitions	18
B. Method Symmetry	24
C. Method Classes: Upper Levels	25
D. Method Classes: Hunters with One Hunt Bell	28
E. Method Classes: Hunters with More than One Hunt Bell	35
5. Method Naming	**37**
A. General	37
B. Class Descriptors	37
C. Naming Constraints	38
D. Variations	40
E. Right to Name	40
F. Exceptions	41
6. Performance Reporting	**42**
A. Definitions	42
B. Content	46
C. Performance Norms	47

7. Record Lengths **49**

A. Requirements 49

B. Notification and Verification 49

C. Reporting 50

8. Method Extension **51**

A. Definitions 51

B. Requirements 52

C. New Extension Processes 53

D. Exceptions 53

9. Related Roles **55**

A. Collections 55

B. Records 55

C. Advice 55

D. Publication 56

E. Analysis and Reporting 56

F. Transitional Arrangements 56

1. Introduction

A. Background

This website houses the Central Council's Framework for method ringing ('the Framework' or 'FMR').

The Framework provides common language and terminology for describing many aspects of method ringing, and it also contains the Central Council's requirements for adding new methods to the Central Council's methods library, for classifying and extending methods, and for record length performances. The Framework is provided to support effective communication within the ringing community, and its use by all is strongly encouraged.

The Framework is a Standard (a type of Policy under the Central Council's rules). It is mandatory for anything included in the CC's various collections and records (methods, variations, calls, extension processes, record lengths), including how things are recorded, classified, analysed and reported. The Framework is implemented by the Central Council's Executive.

B. Material

This website contains both (1) the Framework, and (2) supporting material that assists in the understanding of the Framework, expands upon certain definitions and requirements in the Framework, and further assists ringers in communicating about method ringing.

The top level parts of Sections 1 to 9 of this website are the Framework. 'Top level' means the parts of these sections excluding the expanded material that displays when the '+' buttons are pressed.

The appendices of this website, as well as the expanded parts of Sections 1 to 9, are supporting material. This supporting material is not part of the CC Framework Policy, and is updated by the relevant Central Council team whenever the need arises.

Method ringing terms that are defined in the Framework are capitalised wherever they are used elsewhere in the Framework.

The Framework is a living document. Its aim is to capture the current essence of method ringing practice, and as that practice evolves, so too will the Framework. Developers of software that embeds any aspect of the Framework (such as method classification or method extension) may therefore wish to take into consideration how their software would handle any future changes to the Framework.

C. Overview of the Framework

Section 3 provides definitions of core method ringing terms that are used to describe methods, compositions and performances. The further explanation in Section 3.D.1 includes a diagram that shows how various core terms relate to one another. The Framework doesn't attempt to define all terms related to method ringing – for this, see the glossary of ringing terms at http://jaharrison.me.uk/Ringing/Glossary.

Section 4 describes how methods are classified (i.e. grouped according to common features). Section 4.A.1 links to a diagram that shows the various classes and sub-classes of method.

Section 5 contains the requirements for naming new methods and recording them in the Central Council's Methods Library.

Section 6 gives best practices for reporting method ringing performances (notably reporting in The Ringing World and on BellBoard), with the objective of avoiding ambiguity and making explicit anything of significance that would not reasonably be assumed.

Section 7 states the requirements for a performance to be added to the Central Council's register of record lengths.

Section 8 describes how methods at different stages can be related to one another, and thus may be eligible to share the same method name.

Section 9 describes functions performed by the Central Council and The Ringing World related to the Framework.

D. Framework Principles

See Appendix F for information on the principles that were employed in the design and development of the Framework.

2. Versioning

Version 1

Implemented February 24, 2019 with an effective date of June 1, 2019.

Version 2

Implemented Janury 26, 2022 with an effective date of May 1, 2022.

Version 2 is the current live version of the Framework.

See Appendix G for information on the development of version 2 and the changes that were made.

See Appendix H for information on the ringing community consultation that took place prior to the implementation of version 2.

3. Fundamentals of Method Ringing

A. Stages

1. Stage

A property of several method ringing concepts that indicates the number of bells participating.

Further explanation:

Stage applies to the following method ringing concepts: Rows, Changes, Blocks, Methods and Compositions.

The application of Stage to each of these concepts will be covered below in the 'Further explanation' sections of the definitions of these concepts. In addition, click here[1] for an overview of all aspects of Stage.

2. Stage Name

Names that are given to the different Stages, as follows:

1 = One; 3 = Singles; 5 = Doubles; 7 = Triples; 9 = Caters; 11 = Cinques; 13 = Sextuples; 15 = Septuples; 17 = Octuples.

2 = Two; 4 = Minimus; 6 = Minor; 8 = Major; 10 = Royal; 12 = Maximus; 14 = Fourteen; 16 = Sixteen; 18 = Eighteen.

Odd and even Stages above 18 are named with the written form of the corresponding Stage number, such as Nineteen and Twenty-Two.

B. Rows

1. Row

A sequence of numbered bells in which no bell appears more than once.

The bells are numbered consecutively, starting from number one.

Example:

531246 is a Row with a Stage of Minor (or equivalently, a Minor Row).

Further explanation:

The Stage of a Row is the number of bells in that Row. Click here[2] for an overview of all aspects of Stage.

Bells are usually numbered in descending order of pitch. Both cardinal numbers (e.g. 1, 2, 3 ...) and ordinal numbers (e.g. 1st, 2nd, 3rd ...) are used. When using cardinals, number 1 is used for the bell with the highest pitch, number 2 for the next highest pitch and so on. When using ordinals, bell number 2 becomes 'the 2nd', bell number 3 becomes 'the 3rd', and so on. However, bell number 1 is usually referred to as 'the treble' rather than 'the 1st', and the bell with the lowest pitch is usually referred to as 'the tenor' rather than the corresponding ordinal.

Bells are numbered relative to how many are in use. For example, if a tower has eight bells but only the six with the lowest pitches are in use, these bells are numbered 1 to 6 instead of 3 to 8.

Rows are normally shown in columns so that the path of individual bells can be visualised by looking down the columns. In order to keep the columns aligned, a fixed font can be used with single characters representing each bell. Conventionally '0' (zero) is used to represent the 10th, and upper (or sometimes lower) case letters (excluding 'I', 'O' and 'X') are used for bells 11 to 33 as follows:

[1] https://framework.cccbr.org.uk/edition2/images/edition2/fundamentals/understanding-stage.pdf
[2] https://framework.cccbr.org.uk/edition2/images/edition2/fundamentals/understanding-stage.pdf

'E' (Eleven); 'T' (Twelve); 'A' (Thirteen); 'B' (Fourteen); 'C' (Fifteen); 'D' (Sixteen); 'F' (Seventeen); 'G' (Eighteen); 'H' (Nineteen); 'J' (Twenty); 'K' (Twenty-one); 'L' (Twenty-two); 'M' (Twenty-three); 'N' (Twenty-four); 'P' (Twenty-five); 'Q' (Twenty-six); 'R' (Twenty-seven); 'S' (Twenty-eight); 'U' (Twenty-nine); 'V' (Thirty); 'W' (Thirty-one); 'Y' (Thirty-two); 'Z' (Thirty-three)

Note that what is defined as a Row in this section is sometimes described as a Change in common ringing parlance, and this usage of Change will be seen in method ringing books and articles. The Framework separately uses Change as the transition between two Rows (see Section 3.C) and this distinction between Row and Change is important in defining a number of other method ringing terms used in the Framework. Row is the preferred term wherever possible.

Technical comment:

While this definition of Row refers to a sequence of *bells*, a Row can comprise, at minimum, one bell.

2. Rounds

A Row in which the bells are ordered from the lowest number to the highest number.

Example:

12345678 is Rounds for a Major Row.

3. Place

The position of a bell within a Row.

Example:

In the Row 142536, the 2nd bell is in 3rd's Place.

Further explanation:

Places are numbered starting with 1 for the first bell in the Row, and increasing sequentially for each next bell in the Row. E.g. the bell that is second in a Row is in Place 2.

Each bell in a Row appears in a different Place to all other bells in that Row, and therefore there are the same number of Places in a Row as there are bells in that Row.

Places are often referred to using ordinal numbers in the possessive form, so Place 2 is often '2nd's Place'. 2nd's Place is a reference to the Place the 2nd bell occupies in Rounds.

The bell in 1st's Place is often said to be leading, and the bell in the last Place of the Row is often said to be lying. E.g. in the Row 214365, the 2nd is leading and the 5th is lying.

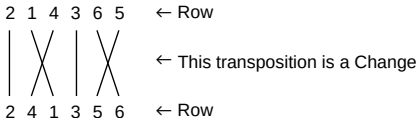
C. Changes

1. Change

A transposition of bells from an existing Row to a new Row, where both Rows have the same Stage.

Example:

A Change might transpose the bells from the Row 214365 to the Row 241356.



Further explanation:

The Stage of a Change is the number of bells on which that Change operates. Click here[3] for an overview of all aspects of Stage.

Since a Change is a transposition of bells, the new Row will comprise the same set of bells as the existing Row.

Note that the same Change applied to different Rows will have the same relative effect. For example, a Change applied to the Row 1342 might produce the Row 1432. The same Change applied to the Row 4123 would produce the Row 4213.

Technical comment:

Note that in mathematics, transposition refers to the exchange of two elements. However in ringing, as well as in more general usage, transposition can involve the exchange of more than two elements. For example, ringers talk about transposing coursing orders, which often involves rotating 3 bells.

2. Identity Change

A Change that transposes each bell to the same Place in a new Row.

Example:

The Change that transposes the bells from Row 54321 to Row 54321 is an Identity Change.

3. Adjacent Change

A Change that transposes each bell to either the same Place or an adjacent Place in a new Row, but which is not an Identity Change.

Example:

The Change that transposes the bells from Row 21436587 to Row 12346857 is an Adjacent Change.

Technical comment:

An Adjacent Change has a minimum Stage of 2.

4. Jump Change

A Change that transposes at least one bell to a different and non-adjacent Place in a new Row.

[3] https://framework.cccbr.org.uk/edition2/images/edition2/fundamentals/understanding-stage.pdf

Example:

The Change that transposes the bells from Row 12345678 to Row 23415678 is a Jump Change because the treble moves from 1st's Place to 4th's Place, which are not adjacent Places.

Further explanation:

When ringing heavy bells, Jump Changes are more commonly used to jump from an earlier Place to a later Place, rather than vice versa, due to the physical difficulty of reducing the time between bell strikes.

Technical comment:

A Jump Change has a minimum Stage of 3.

D. Blocks

1. Block

A sequence of Changes, all with the same Stage, and the Rows produced by applying these Changes, starting from an initial Row.

Example:

	1 2 3 4 5 6	← Initial Row
x 16	2 1 4 3 6 5	
	2 4 1 6 3 5	
x 14	4 2 6 1 5 3	
	4 6 2 1 3 5	
x 16	6 4 1 2 5 3	
	6 1 4 5 2 3	
x 12	1 6 5 4 3 2	
	1 6 4 5 2 3	← Final Row

Further explanation:

The Stage of the Changes in the sequence is the same as the Stage of the initial Row.

A Block has the Stage of its constituent Rows. Click here[4] for an overview of all aspects of Stage.

A sequence of Changes is used to produce a sequence of Rows as follows:

- Apply the first Change in the sequence to an initial Row to produce a second Row.
- Then apply the second Change in the sequence to the second Row to produce a third Row.
- Then apply the third Change in the sequence to the third Row to produce a fourth Row, and so on.
- The Row that is produced by applying the final Change in the sequence is referred to as the final Row.

Since a Block is generated using Changes, which are transpositions of bells, all Rows in a Block will comprise the same set of bells.

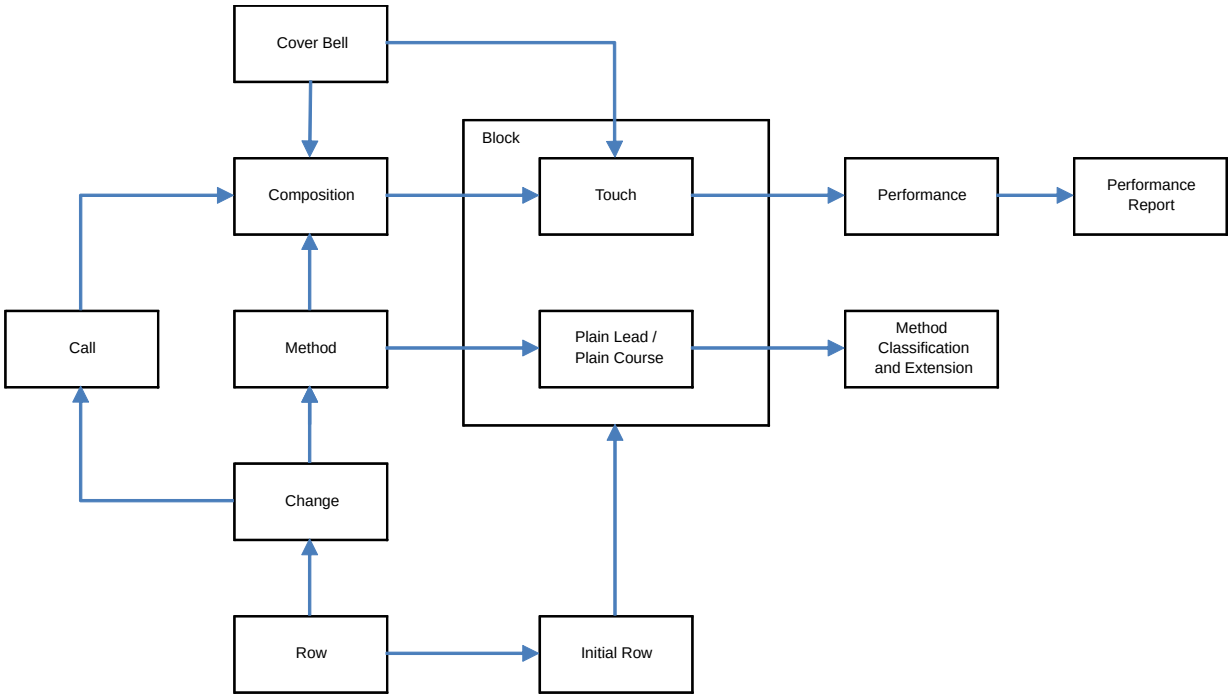
Note that a Block always has one more Row than it has Changes.

See Section 3.E.1 (Method) below for an explanation of the difference between a Method and a Block.

Dividing lines are often drawn in Blocks, such as those shown over the initial and final Rows in the example above. Dividing lines have no effect on the Rows produced – they are merely to highlight features of a Block. In this example, the dividing lines highlight that the treble moves in a symmetrical pattern in the Block.

The following diagram shows the relationships between the various components of method ringing. Terms in the diagram that are not defined above are defined elsewhere in the Framework.

[4] https://framework.cccbr.org.uk/edition2/images/edition2/fundamentals/understanding-stage.pdf



Technical comment:

The term Block is also used in method ringing to refer to a sequence of Changes without reference to the Rows they can produce. For example, a Composition (see Section G below) may specify a Block of Changes that is repeated at different points in the Composition. While the Framework uses the term Block to refer to a sequence of Changes and the Rows produced by applying these Changes, this does not preclude the term Block from being used to refer just to a sequence of Changes in other contexts.

A Block has a minimum size of one Change and two Rows.

Note that since a Composition might specify just to ring a Plain Lead or a Plain Course of a single Method, a Plain Lead and a Plain Course can be Touches as well as Blocks.

2. Round Block

A Block whose final Row is the same as its initial Row.

Example:

	1 2 3 4	← Initial Row
x 14	2 1 4 3	
	2 4 1 3	
x 14	4 2 3 1	
	4 3 2 1	
x 14	3 4 1 2	
	3 1 4 2	
x 14	1 3 2 4	
	1 2 3 4	← Final Row

Further explanation:

In the example above, the final Row is the same as the initial Row, and therefore this is a Round Block.

E. Methods

1. Method

A sequence of Changes all of the same Stage, or a process to generate such a sequence.

Example:

The sequence of Changes x16x14x16x12 is the Method that has been given the name Little Bob Minor.

Further explanation:

A Method has the Stage of its constituent Changes. Click here[5] for an overview of all aspects of Stage.

Individual Changes and sequences of Changes may be represented using place notation, and place notation is used in the example above as well as elsewhere in the Framework. Place notation is described in Appendix A.

A Method is usually referred to by a name it is given. A Method (including its name) may be recorded in the Central Council's Methods Library when certain requirements have been met. These requirements are described in Section 5.

A Method is distinct from a Block in that a Method only defines Changes, not Rows. A Block, on the other hand, defines both Changes and Rows. A single Method can produce many Blocks. For example, if a Method is rung starting from Rounds, this produces a Block. If the same Method is rung starting from a non-Rounds Row (such as 'Queens'), the resulting Block is different from the one produced by starting from Rounds.

2. Static Method

A Method whose sequence of Changes is fixed and finite.

Example:

x16x14x16x12 (Little Bob Minor) is an example of a Static Method. This Method can be represented by a fixed length place notation.

3. Dynamic Method

A Method whose sequence of Changes is not fixed or is not finite.

Example:

Dixon's Bob Minor.

Further explanation:

Dixon's Bob Minor specifies that Rows are produced by successively applying the pair of Changes x16 except (a) if the Treble is leading after an x Change, replace the next 16 Change with a 12 Change, and (b) if the 2nd or 4th is leading after an x Change, replace the next 16 Change with a 14 Change.

In this Method, the sequence of Changes varies depending on the Rows produced, so using a different initial Row could produce a different sequence of Changes. This Method's sequence of Changes is therefore not fixed, and so Dixon's Bob Minor is an example of a Dynamic Method.

Technical comment:

A Dynamic Method should be capable of producing a readily-determinable sequence of Changes. A process such as 'Ring the Changes [3.1.3.1.3.{7|5}]^840 where at each choice of 7 or 5, 7 is chosen unless it is impossible to generate a Round Block of the Extent with any choice of following 7 or 5s' is not a valid Dynamic

[5] https://framework.cccbr.org.uk/edition2/images/edition2/fundamentals/understanding-stage.pdf

Method, even though this is a well-defined process.

Dynamic Methods are in their infancy, and more precise definitions in this area may be developed for future versions of the Framework if there is sufficient interest by the ringing community.

4. Plain Lead

A Block that is produced by applying a Static Method's sequence of Changes once.

Example:

Bastow Little Bob Minimus has the sequence of Changes x12x14. A Plain Lead of this Method, starting from Rounds, is therefore:

	1 2 3 4	← Initial Row
x	2 1 4 3	
12	2 1 3 4	
x	1 2 4 3	
14	1 4 2 3	← Final Row

Further explanation:

Plain in this term means that no Calls are used. Calls are covered below in Section 3.F. When the term Lead is used rather than Plain Lead, this means that Calls may or may not be used. However, it is often clear from the surrounding context whether or not Calls are involved, and so in practice Plain Lead will often be shortened to Lead in method ringing texts, with it understood that no Calls are involved.

5. Plain Course

A Block that is produced by applying a Static Method's sequence of Changes repeatedly, until a Round Block is obtained.

Example:

Bastow Little Bob Minimus has the sequence of Changes x12x14. A Plain Course of this Method, starting from Rounds, is therefore:

	1 2 3 4	← Initial Row
x	2 1 4 3	
12	2 1 3 4	
x	1 2 4 3	
14	1 4 2 3	
x	4 1 3 2	
12	4 1 2 3	
x	1 4 3 2	
14	1 3 4 2	
x	3 1 2 4	
12	3 1 4 2	
x	1 3 2 4	
14	1 2 3 4	← Final Row

Further explanation:

'Plain' in this term means that no Calls are used. Calls are covered below in Section 3.F. When the term Course is used rather than Plain Course, this means that Calls may or may not be used.

In the above example, since it takes three applications of the Method's sequence of Changes to produce a Round Block, the Plain Course of Bastow Little Bob Minimus therefore comprises three Plain Leads.

A Plain Course of a Method is a commonly used Block in method ringing. Unless otherwise specified, a Plain Course is started from Rounds. A Plain Course starting from Rounds is often referred to as *The* Plain Course of a Method, whereas starting from any other Row might be referred to as *A* Plain Course.

Technical comment:

For a Method that is false in the Plain Course (Truth is covered in Section 3.J below) the initial Row might come up in the middle of a Plain Lead. However the Plain Course does not end until the initial Row comes up as the last Row after applying a Static Method's whole sequence of Changes.

While Plain Lead and Plain Course are defined above in terms of a Static Method, these terms may also be applicable to some Dynamic Methods. For example, in Dixon's Bob Minor with Rounds as the initial Row, Rounds comes up again after 64 Changes (with no Calls). This would be considered the Plain Course of Dixon's.

F. Calls

1. Call

An instruction to replace Change(s) from a Method with different Change(s), change the current Method to a different one, or affect Cover Bell(s).

Example:

The last Change in Little Bob Minor's sequence of Changes is '12'. This is usually replaced with the Change '14' when a 'bob' is called. A 'bob' is a commonly-used Call in method ringing.

Further explanation:

A Call is not part of the definition of a Method.

Using Calls to change the current Method to a different one and to affect Cover Bell(s) are covered in Sections 3.G and 3.H respectively, below.

In the application of Calls to replacing Change(s) from a Method with different Change(s), a Call most commonly replaces an individual Change with another, as in the example above. But a Call may also add Changes, remove Changes, or replace a set of Changes with a different set of Changes.

When a Call adds or removes Changes to or from a Method, these Changes are respectively added to or subtracted from the count of Changes for that Method that is used for Performance Reporting purposes. Performance Reporting is covered in Section 6.

Technical comment:

When a Call affects the Changes of more than one Method (i.e. in spliced), the Composition (see Section 3.G below) should make clear how the Change count of each Method is affected (if at all) in order to enable accurate Performance Reporting (see Section 6).

2. Calls Library

Commonly used Calls are recorded in the Central Council Calls Library. This Library also includes Calls that are used to define Variations (see Section 5.D).

Further explanation:

The CC Calls Library is in development.

G. Compositions

1. Composition

An arrangement of Method(s) and Call(s) that produces a sequence of Changes, all with the same Stage.

Further explanation:

A Composition has the Stage of the Changes it produces. Click here[6] for an overview of all aspects of Stage.

Larger Compositions may be created by combining other Compositions. For example, seven different Compositions of 720 Minor Changes may be concatenated to produce a Composition of 5040 Minor Changes.

In addition to using a Method's whole sequence of Changes one or more times in a Composition, any subset of a Method's sequence of Changes may also be used.

When a Composition that uses more than one Method is rung, the instruction to change from one Method to another is referred to as a Call, and this is another application of the term Call.

Technical comment:

A Composition may use two or more Methods side by side to produce Changes with a higher Stage. For example, a Minor Method and a Doubles Method may be combined side by side to create a Composition that produces Cinques Changes (so the Composition has a Stage of Cinques). Similarly, a Composition may combine a Minor Composition and a Doubles Composition side by side to create a Composition that produces Cinques Changes.

2. Spliced

A Composition involving more than one Method is described as Spliced if any changes of Method in the Block produced by the Composition occur at a Row that is not the same as the initial Row of the Block.

Further explanation:

If, for example, a Block starts in Rounds, and changes of Method only occur at Rounds during the Block (such as in a Peal of 7 Minor Methods) then the Composition is not Spliced. But if any change of Method occurs during the Block at any Row other than Rounds, then the Composition is Spliced.

H. Cover Bells

1. Cover Bell

A bell that occupies a Place in a Row that is not one of the Places operated on by the Method(s) of a Composition.

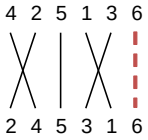
Example:

A Composition for a Doubles Method is used with an initial Row that has a Stage of Minor, and this produces a Minor Block. The Composition operates on Places 1 to 5. 6th's Place therefore contains a Cover Bell (usually the tenor).

Further explanation:

A Cover Bell remains in the same Place from one Row to the next. In the example above, the Doubles Changes of the Composition are deemed to become Minor Changes with a place notation of '6' added to each Change.

[6] https://framework.cccbr.org.uk/edition2/images/edition2/fundamentals/understanding-stage.pdf



In a Row containing one or more Cover Bells, the numbering of the Places may be modified to exclude the Cover Bell(s). In the example above, the bell in 5th's Place (rather that 6th's) may be said to be lying.

A Composition of a given Stage can be used to produce either a Block with the same Stage as the Composition, or a Block with a higher Stage. When the Stage of the Block is higher than the Stage of the Composition, any Places not operated on by the Composition contain Cover Bells.

In the example above, the Doubles Composition is applied to Places 1 to 5 so that 6th's Place contains a Cover Bell. The tenor as Cover Bell is the usual implementation when the Stage of a Block is one higher than the Stage of a Composition. However, the Doubles Composition could be applied to Places 2 to 6 so that 1st's Place contains a Cover Bell.

Since Cover Bells can be generated by using a Composition of a given Stage to produce a Block of a higher Stage, Compositions often do not need to specify the transposition of Cover Bells as these are inferred. However, since the Changes produced by a Composition are all of the same Stage, there are scenarios where a Composition's Changes include the transposition of Cover Bells for some or all Changes.

As example #2, if a Composition uses a Doubles Method initially, applying to Places 1 to 5, and then switches to a Minor Method, applying to Places 1 to 6, then the Composition's Changes are all specified as Minor Changes, with a place notation of '6' added to all Doubles Changes. This Composition therefore has a Stage of Minor. However, even though the Composition has a Stage of Minor, it is still correct to describe the bell in 6th's Place as a Cover Bell when the Doubles Method is being rung, because for these Rows, 6th's Place is not being operated on by the Method of the Composition.

As example #3, a Composition might use a Doubles Method but is designed to produce a Minor Block, and Calls are used that occasionally exchange the Cover Bell in 6th's Place. In this case the Composition's Changes are all specified as Minor Changes (so the Composition's Stage is Minor), with a place notation of '6' added to all Changes where the Cover Bell is not affected. For example, if the Composition's Doubles Method has a '3' Change, this becomes a '36' Change in the Composition. Where the Cover Bell is affected by a Call, what otherwise might have been a '36' Change might instead become a '34' Change.

Click here[7] for an overview of all aspects of Stage.

Technical comment:

As discussed above under Compositions (Section 3.G), it is possible for a Composition to use two or more Methods side by side, or to use two or more Compositions side by side. As a further variation, a Composition might also result in an interior Cover Bell. For example, a Composition might be designed to be used with Maximus Rows, using a Doubles Method in Places 1 to 5 and a Minor Method in Places 7 to 12. 6th's Place does not have any Methods operating on it, and therefore 6th's Place contains a Cover Bell. In this case, the Composition would produce Maximus Changes (and therefore have a Stage of Maximus), and each of its Changes would include the place notation '6'.

Every Row in a Block (except for the initial Row) has at least one Method involved in generating it. A Row is not made up solely of Cover Bells.

2. Variable Cover

A Composition is described as Variable Cover if one or more Cover Bells are affected by any of the Composition's Calls.

Further explanation:

The Composition described in example #3 in Section 3.H.1 above is a Variable Cover Composition.

[7] https://framework.cccbr.org.uk/edition2/images/edition2/fundamentals/understanding-stage.pdf

Note that a Composition is not considered Variable Cover if a Cover Bell is only affected by a change to a higher Stage Method such that the (former) Cover Bell is now one of the bells on which the Method's Changes operate.

I. Touches

1. Touch

A Block that is produced by applying a Composition's sequence of Changes.

Further explanation:

Note that since a Composition can be a Plain Course of a Method, a Plain Course is therefore also a Touch. The same applies to a Plain Lead.

2. Length

The number of Changes in a Touch.

Technical comment:

If a Touch uses side by side Changes to produce a single Row, only one Change per Row is counted when determining Length.

3. Short Touch

A Touch with a Length of 1 to 1249 Changes.

4. Quarter Peal

A Touch with a Length of 1250 to 2499 Changes.

5. Half Peal

A Touch with a Length of 2500 to 4999 Changes.

6. Peal

A Touch with a Length of 5000 or more Changes.

7. Long Length

A Touch with a Length of 10000 or more Changes.

Further explanation:

A Long Length is also a Peal – it is a subset thereof.

8. Date Touch

A Touch with a Length that corresponds to a year being commemorated.

Example:

A Date Touch commemorating the year 2018 would have 2018 Changes and would also be a Quarter Peal.

J. Truth

1. Extent

The complete set of distinct Rows possible for a given set of bells.

Further explanation:

The number of Rows in an Extent for a given set of bells is the factorial of the number of bells in the set. E.g. for 4 bells, the Extent has 4! Rows = 1 * 2 * 3 * 4 = 24 Rows.

When Extent is used on its own, it refers to the unordered set of possible Rows at the Stage in question. However when used with a Method or Composition (e.g. an Extent of Plain Bob Minor), this generally refers to an ordered Extent generated using the Changes of Plain Bob Minor (and any necessary Calls).

Note that usage such as 'Two Extents of Grandsire Doubles' can be ambiguous. This could mean two discrete Extents rung back-to-back, or a Touch of 240 Grandsire Doubles that includes every possible Row exactly twice each. Care should be taken in how this is communicated if the distinction is important.

2. Fixed Place

A given Place that, in every Row of a Touch, is occupied by the same bell.

3. Effective Stage

The Stage of a Touch after any Fixed Places are excluded.

Technical comment:

Note that the Effective Stage of a Block all of whose Rows are the same is zero. An Effective Stage of one is not possible – once the Effective Stage falls below two, it becomes zero. Ringing a Block that has an Effective Stage of zero (such as all Rounds or all Queens) is not considered method ringing.

4. True

A Touch is True if:

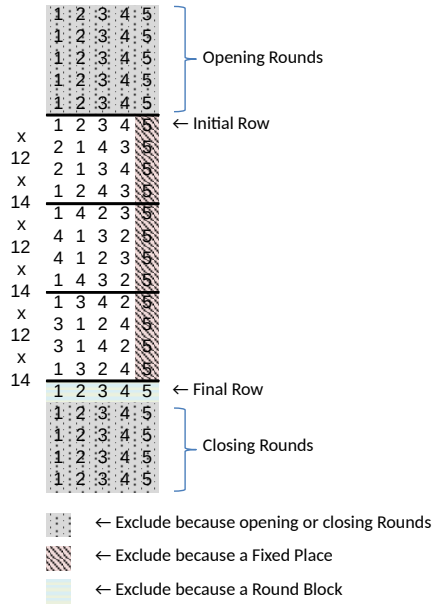
a) It contains zero or more Extents at the Touch's Effective Stage; and

b) Any Rows in addition to the above are distinct.

Further explanation:

If a Touch is a Round Block, the final Row is excluded when determining whether the Touch is True.

When ringing a Touch, it is traditional to ring the initial Row (usually Rounds) several times before starting the Changes, and similarly to ring the final Row several times before standing the bells. These Rows (shown in the diagram below) are excluded when determining Truth.



5. Complete (Incomplete)

A Touch is Complete if it contains all possible Rows at the Touch's Effective Stage exactly once, or exactly the same number of times. (Anything else is Incomplete.)

Further explanation:

A 540 of Minor may be True but is Incomplete;

A 720 of Minor may be True, and if it is, it will also be Complete;

A 1260 of Minor may be True but is Incomplete;

A 1440 of Minor may be True, and if it is, it will also be Complete;

etc.

6. Accepted Truth

A Touch rung on n bells has Accepted Truth if it is True, or it is comprised of Round Blocks that can be divided into two groups where:

a) One group forms a True Touch with an Effective Stage of n and contains at least one Extent at Stage n;

b) The other group forms a True Touch with an Effective Stage of n-1 and the same bell rings in nth's Place in every Row; and

c) Only one group may be an Incomplete Touch at its respective Effective Stage.

Further explanation:

Accepted Truth applies most commonly to Touches of Doubles and Minor rung on 6 bells. When ringing Doubles, the tenor usually rings in 6th's Place as a Cover Bell.

A Row occuring at a change of Effective Stage (which will be the same as the initial Row since changes of Effective Stage can only occur at the end of a Round Block) is counted as a Row of the following Round Block when determining the Truth of the two groups. The initial Row is counted as part of the first Round Block when determining the Truth of the two groups, and the final Row is excluded.

Note that a Touch with Accepted Truth will contain one or more Extents at Stage n, zero or more Extents at Stage n-1, and any remaining Rows will be distinct.

K. Performances

1. Performance

The successful ringing of a Touch where the Band strives to maintain a high standard of ringing, and errors in ringing or calling are corrected quickly.

Further explanation:

Band is defined in Section 5.A.4.

2. Performance Report

A report of a Performance that includes the contents specified in Section 6.

3. Record Length

A Performance of a Long Length (i.e. 10000 or more Changes) that is the longest Length yet rung in a single Method or the same set of Methods.

Further explanation:

A Record Length is also a Peal and a Long Length – it is a subset of both.

Requirements for CCCBR recognition of Record Lengths are described in Section 7.

4. Method Classification

A. Definitions

1. Method Class

A group of Methods that contain the same, defined features. Click here[8] for a diagram that shows the Method Classes and the relationships between them. These classes are defined in Sections 4.C to 4.E below, except that Static Method and Dynamic Method are defined in Section 3.E.

Further explanation:

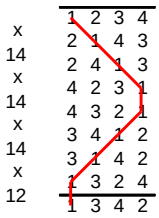
Note that Dynamic Methods are not currently sub-classified to lower levels.

The classification system for method ringing evolved over many decades and as a result is more complex than if we were starting from scratch. Making any large-scale changes to the classification system is problematic because class terms form part of a Method's Title (see Section 5.A.3), so changes to classifications also alter Method Titles, making it harder to interpret historical records of method ringing Performances.

2. Hunt Bell

A bell that ends a Plain Lead in the same Place as it started.

Example:



Further explanation:

The place notation for the Method Plain Bob Minimus is x14x14x14x12. This place notation produces one Plain Lead when applied once to an initial Row. Using an initial Row of 1234 (Rounds), the Plain Lead shown in the diagram is produced. To identify which are the Hunt Bell(s), look at the final Row produced (in this example it is 1342). The Hunt Bell(s) are the bell(s) that are in the same Place(s) in the final Row as they were in the initial Row. In Plain Bob Minimus, the bell in 1st's Place in the initial Row (in this example, the treble) is therefore the only Hunt Bell.

Technical comment:

Note that a Plain Lead is defined in terms of a Static Method. The concept of a Hunt Bell may not apply to Dynamic Methods.

3. Working Bell

A bell that ends a Plain Lead in a different Place from where it started.

Further explanation:

Using the example in Section 4.A.2 above, the bells in 2nd's, 3rd's and 4th's Places in the initial Row (in this example, the 2nd, 3rd and 4th respectively) are Working Bells because at the final Row of a Plain Lead they

[8] https://framework.cccbr.org.uk/edition2/images/edition2/classification/classification-chart.pdf

have not returned to the same Places they were in at the initial Row.

Technical comment:

Note that a Plain Lead is defined in terms of a Static Method. The concept of a Working Bell may not apply to Dynamic Methods.

4. Stationary Bell

A Hunt Bell that occupies the same Place in every Row of a Plain Lead.

Example:

3rd's Place bell in Martyrs Link Little Bob Major[9] is a Stationary Bell.

Technical comment:

Note that a Plain Lead is defined in terms of a Static Method. The concept of a Stationary Bell may not apply to Dynamic Methods.

5. Path

The movements of a bell between its successive Places in the Rows of a Block.

Where a Path ends in the same Place as it started, it is considered to be a loop, and these two Places coincide and are only counted once.

Two such Paths are considered equal if the first Path can be started from a Place that gives the second Path.

Example:

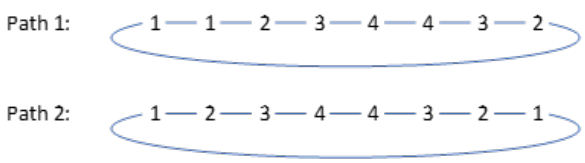
A Path is commonly depicted by drawing a line through the bell in question.

The treble's Path is depicted with a red line in the example in Section 4.A.2 above.

Since this Path starts and ends in the same Place (1st's), it is considered to be a loop, and the Path therefore comprises 8 Places.

Further explanation:

Consider the following two Paths, which are both loops:



The numbers in the diagram above represent the Places through which the Paths pass.

These two Paths are considered to be the same because starting Path 1 from its second Place gives Path 2.

6. Little Path

A Path which does not involve occupying all the Places of the Stage of a Block.

[9] https://complib.org/method/31112

Example:

See the example of a Little Path in Section 4.D.6 below.

Further explanation:

A Little Path is most commonly used in reference to a Hunt Bell. For example, the Hunt Bell Path in Little Bob Minor is a Little Path. However, a Little Path can also be used with Working Bells. For example, 2nd's Place Bell in Bristol Surprise Major[10] can be said to be a Little Path, and all the Working Bells in Magenta Little Place Maximus[11] have Little Paths over the full Plain Course (and its Hunt Bells also all have Little Paths).

Note that a Stationary Bell has a Little Path.

7. Hunting

A Path that is formed by progressing from an earlier Place to a later Place, or vice versa (but not a combination of both), at the rate of one Place per Row. (Alternative form: a bell Hunts.)

Further explanation:

In the diagram in Section 4.A.2 above, the treble Hunts up in the first half of the lead, and then Hunts down in the second half of the lead.

8. Dodging

Moving one Place in the opposite direction in an otherwise Hunting Path. (Alternative form: a bell Dodges.)

Further explanation:

If the Hunting Path 2, 3, 4, 5 is modified to become 2, 3, 4, 3, 4, 5, then the step back in 3-4 is referred to as Dodging. A step back when otherwise hunting up is referred to as Dodging up, and a step back when otherwise hunting down is referred to as Dodging down.

9. Leadend Change

The last Change in a Method's sequence of Changes.

Example:

Little Bob Minor has the sequence of Changes x16x14x16x12. The Leadend Change is the last of these Changes – i.e. 12.

Further explanation:

Also see the diagram in Section 4.A.12 that shows the location of Leadend Changes.

Technical comment:

Note that the concept of a Leadend Change may not apply to Dynamic Methods.

10. Halflead Change

The Change that occurs at the halfway point in a Method's sequence of Changes.

[10] https://complib.org/method/19048
[11] https://complib.org/method/29113

Example:

Little Bob Minor has the sequence of Changes x16x14x16x12. This sequence is 8 Changes in Length. The Halflead Change is therefore the 4th Change in the sequence – i.e. 14.

Further explanation:

Note that Halflead Change is a concept that applies to Methods whose sequence of Changes is even in Length. The Halflead Change occurs at the position that is the sequence Length divided by 2.

Also see the diagram in Section 4.A.12 that shows the location of Halflead Changes.

Technical comment:

Note that the concept of a Halflead Change may not apply to Dynamic Methods.

11. Leadend

When applying a Method's sequence of Changes to produce a Plain Lead, the Leadend is the Row to which the Leadend Change is applied.

Further explanation:

See the diagram in Section 4.A.12 that shows the location of Leadends.

Technical comment:

Note that a Plain Lead is defined in terms of a Static Method. The concept of a Leadend may not apply to Dynamic Methods.

12. Leadhead

When applying a Method's sequence of Changes to produce a Plain Lead, the Leadhead is the Row to which the first Change of the sequence is applied.

Further explanation:

The formal definitions for Leadhead and Leadend are provided above, and this terminology is helpful when, for example, a Composition is being tested for Truth using tables of false course heads.

But note that in practice, ringers and conductors often refer to the Leadhead Row as the lead end.

Example:

The diagram below shows the Leadheads, Leadends, Halflead Changes and Leadend Changes for a Plain Course of Plain Bob Minimus.

		1 2 3 4	Leadhead
	x	2 1 4 3	
	14	2 4 1 3	
	x	4 2 3 1	
Halflead change	14	4 3 2 1	
	x	3 4 1 2	
	14	3 1 4 2	
	x	1 3 2 4	Leadend
Leadend change	12	1 3 4 2	Leadhead
	x	3 1 2 4	
	14	3 2 1 4	
	x	2 3 4 1	
Halflead change	14	2 4 3 1	
	x	4 2 1 3	
	14	4 1 2 3	
	x	1 4 3 2	Leadend
Leadend change	12	1 4 2 3	Leadhead
	x	4 1 3 2	
	14	4 3 1 2	
	x	3 4 2 1	
Halflead change	14	3 2 4 1	
	x	2 3 1 4	
	14	2 1 3 4	
	x	1 2 4 3	Leadend
Leadend change	12	1 2 3 4	

Technical comment:

Note that a Plain Lead is defined in terms of a Static Method. The concept of a Leadhead may not apply to Dynamic Methods.

13. Making one or more Places

Making a Place is occupying the same Place for two consecutive Rows. (Alternative form: a bell Makes a Place.)

Making one or more Places is occupying the same Place for two or more consecutive Rows.

Further explanation:

In the example in Section 4.A.2 above, the treble Makes 4th's at the Halflead Change, and Makes 1st's at the Leadend Change.

Other forms also exist. For example, occupying the same Place for 3 consecutive Rows might be referred to as 'Making a Place twice' or (e.g.) 'Making 3 blows in 2nd's'.

See the example of Making one or more Places in Section 4.D.1.1 below.

14. Making an Internal Place

Making a Place that is neither the first Place nor the last Place of the Block in question. (Alternative form: a bell Makes an Internal Place.)

Further explanation:

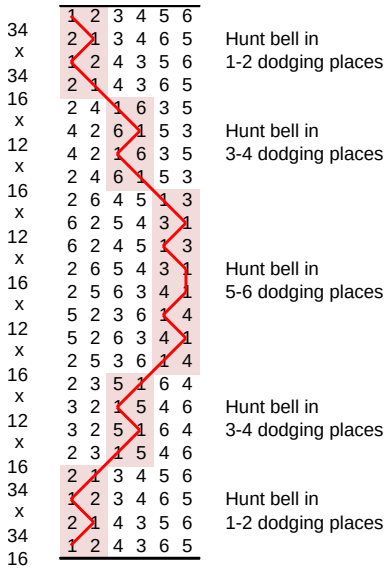
See the examples of Making an Internal Place in Sections 4.D.2.2 and 4.D.2.3 below.

15. Dodging Places

Successive pairs of adjacent Places of the Path of the Hunt Bell in question.

Example:

In a Path where the Hunt Bell moves between 1st's Place and 6th's Place, the Dodging Places are 1-2, 3-4 and 5-6. The diagram below shows the Dodging Places for Kent Treble Bob Minor.



Further explanation:

This term is only used with Paths that span an even number of Places, so the span can always be divided into pairs of Places.

As the term implies, Dodging Places are a pair of Places in which a Dodge can occur. However, note that in some Methods, Dodges can also occur across two Dodging Places, e.g. in 4-5.

16. Cross Section

A Change at which the Hunt Bell in question crosses from one pair of Dodging Places to the next.

Further explanation:

Cross Sections are used in the classification of Treble Dodging Methods – see Sections 4.D.2.1, 4.D.2.2 and 4.D.2.3 below.

In a Method with more than one Hunt Bell, the Cross Sections may be defined by only a subset of the Hunt Bells – see Section 4.E.4 below.

Example:

In a Method with one Hunt Bell that moves between 1st's Place and 6th's Place, since the Dodging Places are 1-2, 3-4 and 5-6, the Cross Sections are the Changes that cause the Hunt Bell to:

1. move from 2nd's Place to 3rd's Place;
2. move from 4th's Place to 5th's Place;
3. move from 5th's Place to 4th's Place; and
4. move from 3rd's Place to 2nd's Place.

17. Rotation

A sequence of Changes can be rotated by considering it as a loop, and starting it at a different point. This process is called Rotation. By using this process on the Changes of a Static Method, other Static Methods can be produced that are Rotations of the original Static Method.

Example:

The Method Plain Bob Minimus has the Changes x14x14x14x12. The Static Method with the Changes x14x12x14x14 is a Rotation of Plain Bob Minimus.

18. Cycle

A Cycle is a set of one or more bells that follow the same Path and return to their starting Places after the same number of Plain Leads. This number of Plain Leads is the same as the number of bells in the Cycle.

Where a Cycle is made up of two or more bells, the bells successively occupy each other's Places at the Leadheads of a Method's Plain Course, and these bells are Working Bells.

Where a Cycle is made up of one bell, this bell is a Hunt Bell.

Further explanation:

The Method Deferential Differential Bob Minor[12] has the Leadheads:

132645, 123564, 132456, 123645, 132564 and 123456.

It can be seen that bells 2 and 3 occupy each other's Places at the Leadheads, and bells 4, 5 and 6 do the same. The treble is a Hunt Bell. This Method therefore has three Cycles: one with one bell, one with two bells, and one with three bells.

B. Method Symmetry

1. Palindromic Symmetry

A Method has Palindromic Symmetry if the same Changes result (after Rotation if needed) when read backwards, that is, when the order of the Changes is inverted.

Further explanation:

Plain Bob Minor has the sequence of Changes x16x16x16x16x16x12. If this sequence is read backwards, the result is 12x16x16x16x16x16x. This can be Rotated to give x16x16x16x16x16x12, which is the same sequence as above. This Method therefore has Palindromic Symmetry.

About 95% of the Methods in the Methods Library (as of 2018) have Palindromic Symmetry.

Example:

Section 4.B.3 below includes an illustration of Palindromic Symmetry.

2. Double Symmetry

A Method has Double Symmetry if the same Changes result (after Rotation if needed) when reversed, that is, when the Places within each Change are inverted.

Further explanation:

Inverting the Places within a Change means to count the Places from the end of the Row rather than the beginning. For example, a Minor Change might have Places made in 1st's and 2nd's, so place notation = 12. 1st's and 2nd's Places counted from the end of a Minor Row are 5th's and 6th's Places. So when the Minor Change 12 is inverted, the result is 56. Similarly, 14 becomes 36, and 34 becomes (in this case, remains) 34.

Double Union Minor has the sequence of Changes x16x36x56x16x14x12. If each of these Changes is inverted, the result is x16x14x12x16x36x56. This can be Rotated to give x16x36x56x16x14x12, which is the same

[12] https://complib.org/method/28182

sequence as above. This Method therefore has Double Symmetry.

About 2% of the Methods in the Methods Library (as of 2018) have Double Symmetry.

Example:

Section 4.B.3 below includes an illustration of Double Symmetry.

3. Rotational Symmetry

A Method has Rotational Symmetry if the same Changes result (after Rotation if needed) when reversed and read backwards.

Further explanation:

Evening Exercise Bob Minor has the sequence of Changes x14x36x16x16x16x16. Inverting these changes (see Section 4.B.2 above) gives: x36x14x16x16x16x16. Reading these Changes backwards gives 16x16x16x16x14x36x. This can be Rotated to give x14x36x16x16x16x16, which is the same sequence as above. This Method therefore has Rotational Symmetry.

About 2% of the Methods in the Methods Library (as of 2018) have Rotational Symmetry.

Note that although the above Method has Rotational Symmetry, it does not also have Palindromic or Double symmetry, even though the process for testing for Rotational Symmetry involves reversing the Changes and reading them backwards.

However, about 93% of the Methods in the Methods Library (as of 2018) that have Rotational Symmetry do also have Palindromic and Double Symmetry. An example of a Method with all three symmetries is Double Court Bob Minor (x14x36x16x36x14x16).

The general rule is that if a Method has any two of the three symmetries (Palindromic, Double and Rotational), it must also have the third. However, a Method can have any one of these three symmetries without having either of the other two. A Method can also have none of the three symmetries.

Example:

Click here[13] for illustrations of the three symmetries.

C. Method Classes: Upper Levels

1. The first three Classes below are based on the Cycles present in a Method. Click here[14] for a table that shows all the possible combinations of Cycles at the Minor Stage, and gives examples of corresponding Methods where available.

2. Hunter

A Static Method that has one or more Hunt Bells.

Example:

[13] https://framework.cccbr.org.uk/edition2/images/edition2/classification/symmetries.pdf
[14] https://framework.cccbr.org.uk/edition2/images/edition2/classification/minor-cycles.pdf

	1 2 3 4
x	2 1 4 3
14	2 4 1 3
x	4 2 3 1
14	4 3 2 1
x	3 4 1 2
14	3 1 4 2
x	1 3 2 4
12	1 3 4 2
x	3 1 2 4
14	3 2 1 4
x	2 3 4 1
14	2 4 3 1
x	4 2 1 3
14	4 1 2 3
x	1 4 3 2
12	1 4 2 3
x	4 1 3 2
14	4 3 1 2
x	3 4 2 1
14	3 2 4 1
x	2 3 1 4
14	2 1 3 4
x	1 2 4 3
12	1 2 3 4

Further explanation:

A Plain Lead of the Minimus Method x14x14x14x12, starting from Rounds, ends with the Row 1342, as shown in the diagram. At the end of one Plain Lead, the treble is the only bell that has returned to the Place it was in at the initial Row (1234), so the treble is this Method's only Hunt Bell. The other 3 bells are therefore Working Bells. Continuing to apply the place notation successively produces the 2 additional leads shown, such that after 3 Plain Leads, Rounds is obtained. This is the Method's Plain Course. Since this Method has at least one Hunt Bell, it is a Hunter. (This Method is Plain Bob Minimus.)

3. Principle

A Static Method that has no Hunt Bells.

Example:

	1 2 3 4
34	2 1 3 4
x	1 2 4 3
34	2 1 4 3
14	2 4 1 3
12	2 4 3 1
14	2 3 4 1
34	3 2 4 1
x	2 3 1 4
34	3 2 1 4
14	3 1 2 4
12	3 1 4 2
14	3 4 1 2
34	4 3 1 2
x	3 4 2 1
34	4 3 2 1
14	4 2 3 1
12	4 2 1 3
14	4 1 2 3
34	1 4 2 3
x	4 1 3 2
34	1 4 3 2
14	1 3 4 2
12	1 3 2 4
14	1 2 3 4

Further explanation:

A Plain Lead of the Minimus Method 34x34.14.12.14, starting from Rounds, ends with the Row 2341, as shown in the diagram. At the end of one Plain Lead, no bell has returned to the Place it was in at the initial Row (1234), so this method has no Hunt Bells. Continuing to apply the place notation successively produces

the 3 additional leads shown, such that after 4 Plain Leads, Rounds is obtained. This is the Method's Plain Course. Since this Method has no Hunt Bells, it is a Principle. (This Method is Stanton Minimus.)

4. Differential

A Static Method in which the Working Bells do not all first return to their starting Places after the same number of Plain Leads.

Example:

	1 2 3 4 5
5	2 1 4 3 5
1	2 4 1 5 3
5	4 2 5 1 3
1	4 5 2 3 1
5	5 4 3 2 1
1	5 3 4 1 2
5	3 5 1 4 2
1	3 1 5 2 4
5	1 3 2 5 4
34	3 1 2 5 4
5	1 3 5 2 4
1	1 5 3 4 2
5	5 1 4 3 2
1	5 4 1 2 3
5	4 5 2 1 3
1	4 2 5 3 1
5	2 4 3 5 1
1	2 3 4 1 5
5	3 2 1 4 5
34	2 3 1 4 5
5	3 2 4 1 5
1	3 4 2 5 1
5	4 3 5 2 1
1	4 5 3 1 2
5	5 4 1 3 2
1	5 1 4 2 3
5	1 5 2 4 3
1	1 2 5 3 4
5	2 1 3 5 4
34	1 2 3 5 4
5	2 1 5 3 4
1	2 5 1 4 3
5	5 2 4 1 3
1	5 4 2 3 1
5	4 5 3 2 1
1	4 3 5 1 2
5	3 4 1 5 2
1	3 1 4 2 5
5	1 3 2 4 5
34	3 1 2 4 5
5	1 3 4 2 5
1	1 4 3 5 2
5	4 1 5 3 2
1	4 5 1 2 3
5	5 4 2 1 3
1	5 2 4 3 1
5	2 5 3 4 1
1	2 3 5 1 4
5	3 2 1 5 4
34	2 3 1 5 4
5	3 2 5 1 4
1	3 5 2 4 1
5	5 3 4 2 1
1	5 4 3 1 2
5	4 5 1 3 2
1	4 1 5 2 3
5	1 4 2 5 3
1	1 2 4 3 5
5	2 1 3 4 5
34	1 2 3 4 5

Further explanation:

A Plain Lead of the Doubles Method 5.1.5.1.5.1.5.1.5.345, starting from rounds, ends with the Row 31254, as

shown in the diagram. At the end of one Plain Lead, no bell has returned to the Place it was in at the initial Row (12345), so this Method has no Hunt Bells. All bells are therefore Working Bells. Continuing to apply the place notation successively produces the 5 additional leads shown, such that after 6 Plain Leads, Rounds is obtained. This is the Method's Plain Course. It will be seen that bells 4 and 5 return to their starting Places at the end of the 2nd, 4th and 6th leads, whereas bells 1, 2 and 3 return to their starting Places at the end of the 3rd and 6th leads. Since the Working Bells do not all first return to their starting Places after the same number of Plain Leads, this is a Differential Method. (This Method is Christ Church Dublin Differential Doubles.)

While the example above shows a Differential with no Hunt Bells, Hunters can also be Differentials. For example, see Deferential Differential Bob Minor[15].

Technical comment:

It takes a minimum of 5 Working Bells to create a Method that is a Differential.

5. Jump Method

A Method that uses one or more Jump Changes.

D. Method Classes: Hunters with One Hunt Bell

1. Plain Method

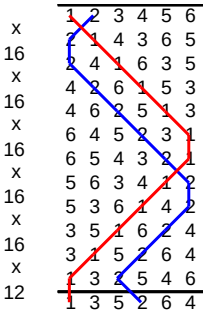
A Hunter in which:

a) The Hunt Bell rings exactly twice in each Place of the Path during a Plain Lead;

b) The Hunt Bell is not a Stationary Bell; and

c) The Method does not use Jump Changes.

Example:



Further explanation:

In this Method, the treble (which is the only hunt bell) rings two times in each Place in a Plain Lead, is not a Stationary Bell, and there are no Jump Changes. This Method is therefore a Plain Method. (This Method is Plain Bob Minor.)

The Path of the treble in the example above is often referred to as a Plain Hunt Path (or Plain Hunting Path) – i.e. Hunt up, Make a Place, Hunt Down, and make another Place. This Path is the only one (for Methods with a single Hunt Bell, and when excluding Jump Changes and Stationary Bells) that meets the criterion of ringing exactly twice in each Place of the Path during a Plain Lead.

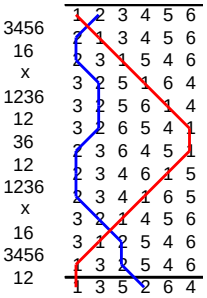
Note that a Plain Path is symmetrical about the two Places Made, and a Plain Path can span either an even or odd number of Places.

[15] https://complib.org/method/28182

1.1. Place Method

A Plain Method in which the Paths of all the bells consist only of Hunting and Making Places, and in which a change in the direction of Hunting is separated by Making one or more Places.

Example:



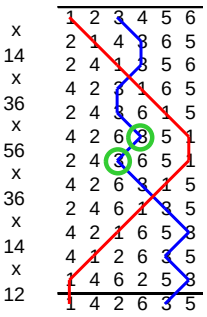
Further explanation:

In this Plain Method, changes in the direction of Hunting by all bells are always separated by Making one or more Places. This Method is therefore a Place Method. (This Method is Foti Place Minor.)

1.2. Bob Method

A Plain Method that is not a Place Method.

Example:



Further explanation:

The 'points' in the blue line of this Plain Method, such as those circled in green, show that a Place is not Made between changes in the direction of Hunting. This Method is therefore a Bob Method. (This Method is Double Oxford Bob Minor.)

2. Treble Dodging Method

A Hunter in which:

a) The Hunt Bell rings more than twice in each Place of the Path during a Plain Lead;

b) The Hunt Bell rings the same number of times in each Place of the Path during a Plain Lead;

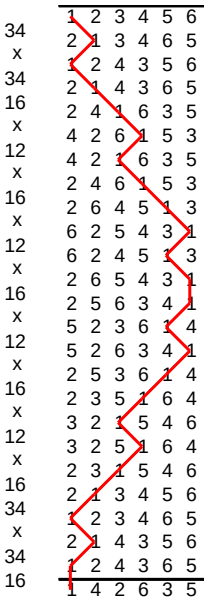
c) The Hunt Bell Makes a Place exactly twice during a Plain Lead;

d) The Path of the Hunt Bell is the same if it is rung backwards;

e) The Hunt Bell is not a Stationary Bell; and

f) The Method does not use Jump Changes.

Example:



Further explanation:

In this Method, the treble (which is the only hunt bell) rings four times in each Place in a Plain Lead. The treble Makes a Place twice during a Plain Lead, once in 6th's and once in 1st's, it is not a Stationary Bell, and there are no Jump Changes. This Method is therefore a Treble Dodging Method. (This Method is Kent Treble Bob Minor.)

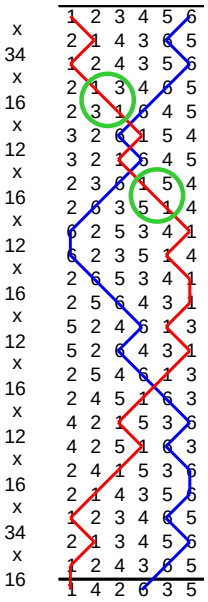
While the above Path is by far the most common Treble Dodging Path, if the single Dodges are all replaced with the same, greater number of Dodges (e.g. double or triple Dodging, etc), the resulting Path is still a Treble Dodging Path.

Note that a Treble Dodging Path is symmetrical about the two Place Made. A Treble Dodging Path can only span an even number of Places.

2.1. Treble Bob Method

A Treble Dodging Method in which no bell Makes an Internal Place at any Cross Section, or which does not have any Cross Sections.

Example:



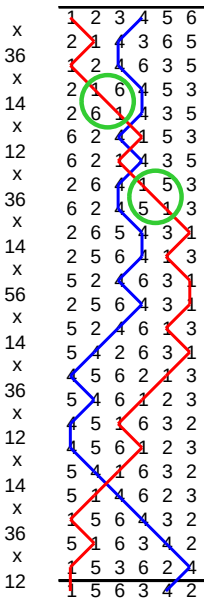
Further explanation:

At this Treble Dodging Method's Cross Sections, which are circled in green, there are no Internal Places Made. The Places Made at both Cross Sections are 1st's and 6th's, which are external Places. Since this Method has Palindromic Symmetry (see Section 4.B.1 above) there is no need to check the Cross Sections in the second half of the lead as these will have the same Places Made as in the first half of the lead. However for non-Palindromic Methods, all Cross Sections must be considered when further classifying a Treble Dodging method. This Method is therefore a Treble Bob Method. (This Method is Oxford Treble Bob Minor.)

2.2. Surprise Method

A Treble Dodging Method in which one or more bells Make an Internal Place at every Cross Section.

Example:



Further explanation:

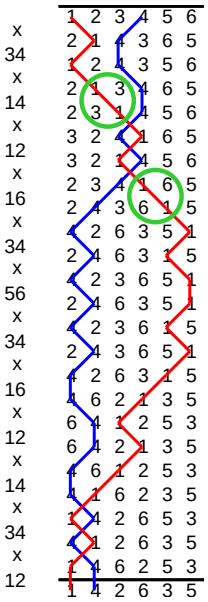
An Internal Place is Made at both of this Treble Dodging Method's Cross Sections, which are circled in green. 4th's is Made at the 2-3 Cross Section, and 3rd's is Made at the 4-5 Cross Section. (See the note about Palindromic Symmetry in Section 4.D.2.1 above, which also applies here.) This Method is therefore a Surprise

Method. (This Method is Cambridge Surprise Minor.)

2.3. Delight Method

A Treble Dodging Method that is neither a Treble Bob Method nor a Surprise Method.

Example:



Further explanation:

An Internal Place is Made at the 2-3 Cross Section of this Treble Dodging Method, but an Internal Place is not made at the 4-5 Cross Section (the Cross Sections are circled in green). (See the note about Palindromic Symmetry in Section 4.D.2.1 above, which also applies here.) Since Internal Places are Made at some but not all Cross Section Changes, this Method is therefore a Delight Method. (This Method is College Bob IV Delight Minor.)

3. Treble Place Method

A Hunter in which:

a) The Hunt Bell rings the same number of times in each Place of the Path during a Plain Lead;

b) The Hunt Bell Makes a Place more than twice during a Plain Lead;

c) The Path of the Hunt Bell is the same if it is rung backwards; and

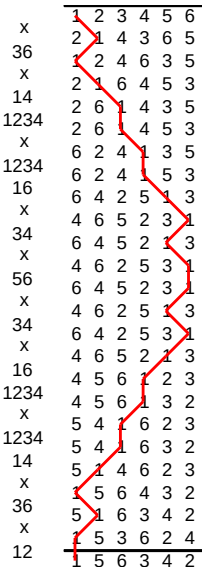
d) The Method does not use Jump Changes;

Or:

a) The Hunt Bell is a Stationary Bell; and

b) The Method does not use Jump Changes.

Example:



Further explanation:

This Method is Stoke Albany Treble Place Minor. Other examples are: Primrose Treble Place Minor[16] and Alderney Treble Place Minor[17].

Note that a Treble Place Path can span either an even or odd number of Places.

4. Alliance Method

A Hunter in which:

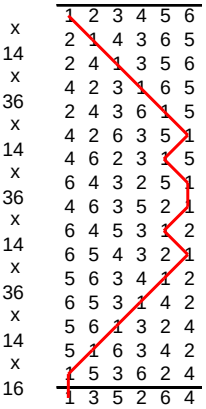
a) The Hunt Bell does not ring the same number of times in each Place of the Path during a Plain Lead;

b) The Path of the Hunt Bell is the same if it is rung backwards;

c) The Hunt Bell is not a Stationary Bell; and

d) The Method does not use Jump Changes.

Example:



Further explanation:

This Method is Alperton Alliance Minor. Another example is: Semiquincentenary Alliance Major[18].

Note that an Alliance Path can span either an even or odd number of Places.

[16] https://complib.org/method/27309
[17] https://complib.org/method/27502
[18] https://complib.org/method/26928

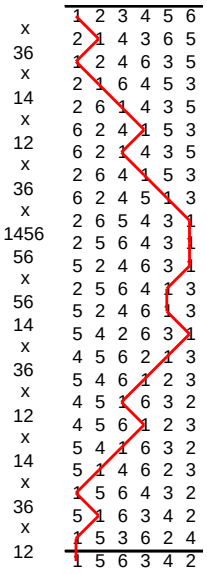
5. Hybrid Method

A Hunter in which:

a) The Path of the Hunt Bell is not Plain, Treble Dodging, Treble Place or Alliance; and

b) The Method does not use Jump Changes.

Example:



Further explanation:

This Method is Seavington St Mary Hybrid Minor.

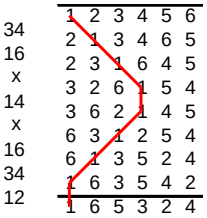
Note that the Path of the Hunt Bell in a Hybrid Method with a single Hunt Bell is asymmetric.

A Hybrid Path can span either an even or odd number of Places.

6. Little Method

A Hunter in which the Hunt Bell has a Little Path.

Example:



Further explanation:

This Method is Canterbury Little Bob Minor. Since the treble, which is the only Hunt Bell, only rings in the first four Places, whereas the Stage of the Method is six, this is a Little Method.

E. Method Classes: Hunters with More than One Hunt Bell

1. If a Hunter has more than one Hunt Bell, it has a further classification of the first Method Class found, when inspecting each of the Hunt Bells, using the order: Plain Method, Treble Dodging Method, Treble Place Method, Alliance Method, Hybrid Method. If none of the Method Classes above are found, the Hunter has no further classification.

Example:

Didymous Delight Major[19].

Further explanation:

Didymous Delight Major has three Hunt Bells in 1st's, 2nd's and 3rd's Places. The treble has a Treble Dodging Path, and the 2nd and 3rd have Treble Place Paths. Since Treble Dodging comes before Treble Place in the order above, this Method has a Treble Dodging classification (and is further classified as Delight as described in Section 4.E.4 below).

Technical comment:

Note that in a Method with two or more Hunt Bells, it's possible for a Hybrid Path to be symmetric (unlike for single Hunt Bell Methods where a Hybrid Path is always asymmetric). An example of this is a Method in which a pair of Hunt Bells dodge together for the entire Plain Lead. Provided the Lead Length is not 4 Changes, these Hunt Bells classify as Hybrid. If the Lead Length is 4 Changes, these Hunt Bells classify as Plain.

2. If the Path(s) of the Hunt Bell(s) of the first Method Class found in Section 4.E.1 above, if any, all meet the definition of a Little Method per Section 4.D.6 above, then the Hunter is classified as a Little Method. If no Method Class was found in Section 4.E.1 above, then if the Paths of all the Hunt Bells meet the definition of a Little Method per Section 4.D.6 above, then the Hunter is classified as a Little Method.

Example:

Seven Stars Differential Little Hybrid Major[20] and Bushey Treble Place Triples[21].

Further explanation:

Seven Stars Differential Little Hybrid Major has two Hunt bells in 5th's and 6th's Places. Both these Hunt Bells have Hybrid Paths, so the Method has a Hybrid classification. These Hunt Bells also both have Little Paths. Since all the Hybrid Paths are Little, this Method is also classified as Little.

Bushey Treble Place Triples has two Hunt bells in 1st's and 2nd's Places. Both these Hunt Bells have Treble Place Paths, so the Method has a Treble Place classification. The 2nd has a Little Path, but the treble does not have a Little Path. Since the Treble Place Paths are not all Little, this Method is not classified as Little.

3. If the first Method Class found in Section 4.E.1 above is a Plain Method, the Method is further classified in accordance with Sections 4.D.1.1 and 4.D.1.2 above.

Example:

Reverse Canterbury Pleasure Place Minor[22] and Glastonbury Bob Triples[23].

Further explanation:

[19] https://complib.org/method/25556
[20] https://complib.org/method/28544
[21] https://complib.org/method/27681
[22] https://complib.org/method/11943
[23] https://complib.org/method/32106

Reverse Canterbury Pleasure Place Minor has two Hunt bells in 1st's and 2nd's Places. Both these Hunt Bells have Plain Paths, so the Method has a Plain classification. Since the Paths of all the bells in this Method consist only of Hunting and Making Places, with changes in the direction of Hunting separated by the Making of one or more Places, this Method is further classified as a Place Method.

Glastonbury Bob Triples has two Hunt bells in 1st's and 2nd's Places. The treble has a Plain Path and the 2nd has a Little Hybrid Path, so the Method has a Plain classification (because Plain comes before Hybrid in the order in Section 4.E.1). Since this Method has Paths that are not limited to Hunting and Making Places (i.e. there are dodges), this Method is further classified as a Bob Method.

4. If the first Method Class found in Section 4.E.1 above is a Treble Dodging Method, the Method is further classified in accordance with Sections 4.D.2.1, 4.D.2.2 and 4.D.2.3 above, using the following:
   a) If the Method has more than one Treble Dodging Hunt Bell, none of which are Little, or all of which are Little, the Cross Section Changes are all the Changes that are a Cross Section Change for any Treble Dodging Hunt Bell.
   b) If the Method has both Little Treble Dodging Hunt Bell(s) and non-Little Treble Dodging Hunt Bell(s), the Cross Section Changes are all the Changes that are a Cross Section Change for any non-Little Treble Dodging Hunt Bell.

Example:

Vincula Surprise Royal[24].

Further explanation:

Vincula Surprise Royal has three Hunt bells in 1st's, 2nd's and 3rd's Places. All these Hunt Bells have Treble Dodging Paths, so the Method has a Treble Dodging classification. None of the Hunt Bells has a Little Path, so the Cross Section Changes are all the Changes that are Cross Section Changes for any of the three Hunt Bells. It can therefore be seen that every 4th change in this Method is a Cross Section Change. Since all of these Changes (16, 18, 30, 50, 70, and 14 at the lead end) include an Internal Place being made, this Method is further classified as a Surprise Method.

[24] https://complib.org/method/22632

5. Method Naming

A. General

1. Methods Library

The full title is the Central Council Methods Library. This is the definitive record of Methods that have been named under the Framework or under earlier Central Council Decisions.

Further explanation:

The Methods Library can be accessed here[25].

2. Method Name

The name given to a Method as recorded in the Methods Library.

3. Method Title

Methods recorded in the Methods Library have a unique tripartite Method Title consisting of: Method Name, Class Descriptor, Stage Name.

Further explanation:

The Class Descriptor is defined in Section 5.B below. For some Methods, the Class Descriptor comprises more than one term. For other Methods, notably Principles, the Class Descriptor may be blank.

4. Band

One or more human ringers ringing some or all of the bells (or simulated bells) in a Performance.

1. Methods recorded in the Methods Library may not have Names that conflict with the constraints described in Section 5.C.
2. Methods recorded in the Methods Library that have in the past been known by a different Name and/or Title will have this recorded, where details are known.

Further explanation:

Requests to add alternative Names / Titles for Methods to the Methods Library should be emailed to methods@cccbr.org.uk. Documentation that supports or demonstrates a reasonably widespread use of the alternative Name / Title should be included with the request.

B. Class Descriptors

1. Class Descriptor

A Method's Class Descriptor is formed by concatenating applicable terms as follows, and in the order shown:

a) Add 'Jump' if the Method is a member of this Class;

b) Add 'Differential' if the Method is a member of this Class;

c) Add 'Little' if the Method is a member of this Class, and if the Method is also a member of one of the

[25] https://methods.cccbr.org.uk

Classes in d) below;

d) Add one of the following if the Method is a member of this Class: ‘Place’, ‘Bob’, ‘Treble Bob’, ‘Surprise’, ‘Delight’, ‘Treble Place’, ‘Alliance’;

e) Otherwise leave blank.

C. Naming Constraints

1. The following constraints apply to a new Method to be added to the Methods Library.
2. A Method Name should follow the Central Council’s requirements on character usage, and should not exceed the maximum number of characters set by the Council.

Further explanation:

The Central Council’s requirements for Method Name character usage and length are described in Appendix B.

3. A new Method may not use the same Name as a Method already recorded in the Methods Library, or a Variation already recorded in the Variations Library (see Section 5.D below), if that would result in their Method Titles being the same.

Further explanation:

This can be more complex than first meets the eye. For example, is ’London No. 3 Surprise Royal’ considered the same as ’London No 3 Surprise Royal’. The Central Council’s Method Name uniqueness requirements are described in Appendix B (Method Name Syntax).

4. The Name given to a new Method that is recorded in the Methods Library may not include Class Descriptor term(s) such that the resulting Method Title is ambiguous as to the Class of the Method.

Example:

Naming a Bob Method ’Quavo Treble’ would give a misleading Method Title, e.g. ’Quavo Treble Bob Major’. It can’t be determined from the Method Title that this Method is a Bob Method and not a Treble Bob Method.

5. Rotations of an existing Static Method are not normally separately named in the Methods Library. However, this is not mandated as there are a few cases where it is beneficial to record a separate Name for a Rotation.

Further explanation:

Rotation is defined in Section 4.A.17. When using a Static Method’s sequence of Changes to produce Rows, the Changes may be used starting and finishing at any point in the sequence. For this reason, a Rotation of an existing Static Method is not usually separately named, because the same Changes can be produced by starting and finishing the existing Static Method at a different point in its sequence of Changes.

An example where separate naming of Rotations is beneficial is for twin hunt methods like Grandsire Triples. When Rotated so that the second Hunt Bell is the treble, and rung with Grandsire Calls affecting the first Hunt Bell rather than the second, it seems like a different method. This Rotation has been rung many times under the name New Grandsire (7.1.7.1.7.1.7.1.7.1.7.1.3.1) and has also been spliced with Grandsire. It is therefore beneficial to include New Grandsire in the Methods Library, and when splicing it with Grandsire, there is a clear way to announce changes of Method.

6. A Static Method normally has a sequence of Changes that is not divisible into two or more equal parts. However this is not mandated as there may be cases where it is beneficial for a Method to have a repeated sequence to give it a different classification. Furthermore, a Method is not normally separately recorded in the Methods Library if its sequence of Changes is a multiple of another Method's sequence of Changes (or a fraction of another Method's sequence of Changes if that other Method has a repeated sequence).

Further explanation:

The method Magenta Little Place Maximus (56.1T.56.1T.56.1T.56.1T.56.1T) could have been recorded in the Methods Library as Magenta Differential Maximus (56.1T) but the band that rang it preferred it to be classified as a Hunter. Subsequently 56.1T and any other repeat of it would not normally be named separately in the Methods Library unless a band felt there was a compelling reason to do so.

7. Method names should not contain words that mislead as to the construction of the Method.

Further explanation:

Certain terms including 'Reverse', 'Single' and 'Double' have been used to prefix Method Names when certain Method features are present. Use of these terms is optional, but if used, they should follow the requirements in a) - c) below. Similarly, terms such as 'True', 'Symmetric' and 'Palindromic' should not be used in Method Names where these could mislead.

a) A Method Name should only begin with 'Reverse' if:

   i) Another Method in the Methods Library has the same Method Title (after excluding 'Reverse' in the new Method); and

   ii) That other Method's Changes, when Rotated if needed, are the Reverse of the new Method's – that is, the Places within each Change are inverted. See Section 4.B.2 for an explanation of inverting the Places within a Change.

b) For non-Little Plain Methods, 'Double' and 'Single' may be used as Method Name prefixes for two Plain Methods that otherwise have the same Method Title where:

   i) Both Methods have the same place notation above the Hunt Bell (or two or more Coursing Hunt Bells – see Appendix D.A.2);

   ii) The Method with the 'Double' prefix is a Double Method (that is, when the Places of each Change are inverted, and then Rotated as needed, the same Method is obtained); and

   iii) The Method with the 'Single' prefix has no Internal Places (see Section 4.A.14) made below the Hunt Bell(s).

c) For Methods not covered by b) above, 'Double' should only be used as a prefix if the Method is a Double Method (as defined in b) above), and 'Single' should only be used as a prefix if the Method is not a Double Method.

Example:

St Simon's Bob Doubles has the place notation 5.1.5.3.5.3.5.1.5.125.

If these Changes are inverted, the result is the sequence 1.5.1.3.1.3.1.5.1.145.

This sequence may be Rotated to give 3.1.5.1.145.1.5.1.3.1.

This resulting Method has been appropriately named Reverse St Simon's Bob Doubles in the Methods Library.

Double Court Bob Minor has the place notation x14x36x16x36x14x16.

If these Changes are inverted, the result is x36x14x16x14x36x16.

This can be rotated to give the same Changes started with.

It is therefore a Double method.

Single Court Bob Minor has the place notation x14x16x16x16x14x16.

Since (1) the Places above the Hunt Bell in Double Court Bob Minor and Single Court Bob Minor are the same, (2) Double Court Bob Minor is a Double Method, and (3) Single Court Bob Minor has no Internal Places made below the Hunt Bell, the Double and Single prefixing of the otherwise common Method Title of Court Bob Minor is appropriate naming.

8. A Method should only be given the same Name as another Method in the Methods Library that has the same Class Descriptor but a different Stage if the requirements for Method Extension are met.

Further explanation:

Method Extension requirements are described in Section 8.

D. Variations

1. A Variation is a named combination of a Method and a Call or Calls.
2. Variations are recorded in the Central Council Variations Library, which cross-references each Variation to the related entries in the Methods Library and Calls Library.

Further explanation:

The CC Variations Library is in development.

3. The Central Council's Variations Library currently only records Plain Doubles Variations.
4. Each Variation recorded in the Central Council's Variations Library:
   a) Has the same Classes as the underlying Method on which it is based, but does not use a Class Descriptor in a Performance Report (see Section 6);
   b) Has a Name that is unique across both Methods and Variations with no Class Descriptor; and
   c) Satisfies the same naming constraints as the underlying Method (see Section 5.C).
5. Variations are incorporated into Compositions, including Spliced and Variable Cover Compositions, in the same way as Methods.

E. Right to Name

1. An unnamed Method or Variation may be named by the Band that first rang it in a Performance that is reported by the Ringing World, providing:
   a) In the case of a Method, the Performance was at least a Quarter Peal in Length or contained an Extent of the Method;
   b) In the case of a Variation, the Performance contained an Extent of the Variation;
   c) The Performance was a Round Block that started and ended in Rounds;
   d) The Performance was a True Touch, or a Touch with Accepted Truth;
   e) The Name does not conflict with any of the requirements of this Section 5 (Method Naming) or Section 8 (Method Extension);
   f) The Composition used is included in the Performance Report; and
   g) The Performance was rung by an all-human Band.

Further explanation:

'Reported by the Ringing World' includes in print and/or on BellBoard.

For item d), note that Accepted Truth encompasses True. See Section 3.J for definitions of these terms.

When naming a Variation, the Stage of the Extent is the same as the Stage of the Variation. The Rows of the Extent may appear anywhere in the Performance - the Extent of the Variation does not have to be rung as a contiguous Block.

For item f), the Composition should be added to the published Performance on BellBoard, but does not need to be published in The Ringing World journal.

Compositions may be described in any form that makes it clear what was rung. E.g. '2 extents each called WHW * 3' for a 1440 of Treble Dodging Minor, or by reference to Composition Library or other online or printed collection.

When naming a Method, it is expected that the Performance contained enough Changes to define the Method uniquely - at least one Plain Lead. However, there may be circumstances where a Band can justify naming a new Method when not all its Changes have been rung, therefore the Framework does not specify a minimum.

Note that any Ringing Style (see Section 6.A.4) can be used in a Performance to name a new Method.

2. The Central Council reserves the right to enter a different name in the Methods Library from the one proposed by a band for a new Method / Variation, or to leave the Method / Variation unnamed, if it considers this necessary.

Further explanation:

This could include considering that the proposed Name is misleading or offensive, or that the naming of an Extension is inappropriate.

F. Exceptions

1. By convention, the Class Descriptor is omitted from the Method Titles of the following Methods that are classified as either Bob, Little Bob or Place Methods:
   a) Grandsire, Double Grandsire, Reverse Grandsire and Little Grandsire, at all Stages;
   b) Union, Double Union and Reverse Union, at Stages of Minor and higher.
2. The above Method Names may not be reused for new Bob Methods, new Little Bob Methods, new Place Methods, or new Methods with no Class Descriptor.
3. The Method known as Little Bob has no Method Name since Little Bob is its Class Descriptor. It will continue to be known as Little Bob.
4. The Method Name 'Plain', which is the most likely name for Little Bob if it had one, may not be used for any new Little Bob Method.

6. Performance Reporting

A. Definitions

1. Performance Title

A Performance Title takes the following form:

a) If the Performance was in a single Method / Variation:

   1. The number of Changes in the Performance;
   2. 'Variable Cover' if the Performance included a Variable Cover Composition (see Section 3.H.2);
   3. The Name of the Method / Variation rung;
   4. The Class Descriptor (if any) of the Method / Variation rung;
   5. The Stage Name of the Method / Variation rung.

b) If the Performance was in more than one Method / Variation:

   1. The number of Changes in the Performance;
   2. 'Variable Cover' if the Performance was of a Variable Cover Composition (see Section 3.H.2);
   3. 'Spliced' if the Performance was of a Spliced Composition (see Section 3.G.2);
   4. Optionally, 'Mixed' if the Performance included more than one Method / Variation, but was not of a Spliced Composition;
   5. The components of the Class Descriptors of the Methods rung that are common to all Methods in the Performance, if any, stated in the same order as given in 5.B.1;
   6. Optionally, 'Plain' can be included if the Methods / Variations rung in the Performance do not all have the same Class Descriptor, but are all members of the Plain Class of Methods. 'Plain' is included in the same position as Bob and Place in 5.B.1;
   7. Optionally, 'Treble Dodging' can be included if the Methods rung in the Performance do not all have the same Class Descriptor, but are all members of the Treble Dodging Class of Methods. 'Treble Dodging' is included in the same position as Treble Bob, Surprise and Delight in 5.B.1;
   8. The Stage Name(s) of the Methods / Variations rung in the Performance, listed in order of lowest Stage to highest Stage, with a comma inserted between Stage Names, except for the last pair of Stage Names which instead have 'and' inserted between them;
   9. Optionally, and in round brackets, the number of Methods / Variations rung in the Performance (if not included here, this information goes at the beginning of the Performance Detail – see Section 6.A.2).

c) In order for Performance Titles not to mislead, if the highest Stage of Method / Variation rung in the Performance exceeds the Effective Stage of the Performance, then:

   1. Layout b) above is always used, even for a single Method / Variation Performance;
   2. The Effective Stage Name is used for b) 8 above, instead of the Stage Name(s) of the Method(s) / Variation(s) rung.

Example:

See Section 6.A.2 below for examples of Performance Titles.

2. Performance Detail

The Performance Detail takes the following form:

a) If the Performance was in a single Method / Variation and the Effective Stage of the Performance was the same as the Stage of the Method / Variation, then the Performance Detail is left blank.

b) If the Performance was in more than one Method / Variation, or was in a single Method / Variation where the Effective Stage of the Performance was less than the Stage of the Method / Variation:

   1. The number of Methods / Variations rung in the Performance if this was not included at the end of the Performance Title;
   2. The number of Changes rung of each Method / Variation, together with the Name and, if not apparent from the Performance Title, the Class Descriptor and Stage of each Method / Variation;
   3. The number of Changes of Method / Variation ('com', 'cov', or 'com/v') in the Performance;
   4. Optionally, key features of the Composition such as 'atw' for 'all the work', and 'eld' for 'each lead different'.

c) Where a multi-Method Performance comprised multiple Round Blocks where each Round Block was a whole multiple of an Extent in Length (e.g. a Peal of 7 single Extent Round Blocks of Minor), the Performance Detail may optionally, and instead of b) 1 to 3 above, list the number(s) of the Extents in the order they were rung, and the Methods / Variations that were rung in each Round Block.

d) As an optional alternative to using the full Class Descriptor in the Performance Detail, the following abbreviations may be used instead:

   - Diff = Differential
   - L = Little
   - B = Bob
   - P = Place
   - S = Surprise
   - D = Delight
   - TB = Treble Bob
   - TP = Treble Place
   - A = Alliance
   - J = Jump

The above are combined as needed – e.g. a Little Alliance Method is LA, a Differential Treble Place Method is Diff TP, etc.

Example:

Following are some examples of Performance Titles and Performance Details:

- Performance Title: 1260 Plain Bob Minor
- Performance Detail: [None required]
- This is the most straightforward scenario, where a single Method was rung in a Performance.
- Performance Title: 1251 Variable Cover Stedman Triples
- Performance Detail: [None required]
- Here a single Method was rung in a Performance, and the Performance included a Variable Cover Composition.
- Performance Title: 5024 Spliced Surprise Major
- Performance Detail: 8m: 640 each Bristol, Pudsey, Rutland, Superlative, Yorkshire; 608 each Cambridge, Lincolnshire, London; 120 com; atw

- Since all the Methods rung in this Performance have a Surprise Class Descriptor, Surprise is included in the Performance Title and the Performance Detail only needs to include the Method Names.
- Performance Title: 5040 Spliced Surprise Minor
- Performance Detail: 9m: (1) Carlisle; (2-3) Beverley, Surfleet; (4) London, Wells; (5) York, Durham; (6) Norwich; (7) Cambridge
- This is an example of a Performance comprising multiple Round Blocks where each Round Block was a whole multiple of an Extent in Length. Since the Performance was of Minor, the first Round Block was 720 Changes, the second Round Block was 1440 Changes, and the remaining Round Blocks were all 720 Changes.
- Performance Title: 1280 Spliced Treble Dodging Major
- Performance Detail: 3m: 416 each Cambridge Surprise, Megan Delight; 448 Imperial Treble Bob; 2 com; atw
- In this example the Methods rung do not have the same Class Descriptor, but they are all Treble Dodging Methods. Treble Dodging may therefore optionally be included in the Performance Title (or otherwise the Performance Title is 1280 Spliced Major), and the Method Names and Class Descriptors are included in the Performance Detail.
- Performance Title: 5000 Spliced Royal (8m)
- Performance Detail: 800 each Dr No Diff S, Kananga S, Zorin S; 640 Largo A; 600 Jaws LA; 560 each Drax LA, Elektra A; 240 Nick Nack Diff; 139 com; atw
- In this example the Methods rung do not have the same Class Descriptor, and they are not all either Plain or Treble Dodging Methods. Therefore no Class Descriptor is used in the Performance Title, and the Method Names and Class Descriptors are included in the Performance Detail. In this example, abbreviations for Class Descriptors have been used (Diff S for Differential Surprise, LA for Little Alliance, etc), and the number of methods rung has been included in round brackets at the end of the Performance Title, rather than at the beginning of the Performance Detail.
- Performance Title: 1320 Mixed Plain Doubles
- Performance Detail: 6m/v: (1-2) Grandsire; (3-4) St Simon's Bob; (5-6) St Martin's Bob; (7) April Day; (8) Kennington; (9-11) Plain Bob
- This example comprises 4 Methods and 2 Variations. The two Variations, April Day and Kennington, are based on Plain Bob, and so are members of the Plain Class. Variations do not use Class Descriptors, so this Performance doesn't have a common Class Descriptor that can be used in the Performance Title. However, Plain may optionally be included in the Performance Title since all Methods and Variations rung are members of this Class. Mixed has also optionally been used in the Performance Title since this is a multi-Method Performance, but doesn't include a Spliced Composition. (Given the optional terms, this Performance Title could be any of: 1320 Mixed Plain Doubles, 1320 Mixed Doubles, 1320 Plain Doubles, or 1320 Doubles.)
- Performance Title: 5040 Plain Doubles and Minor
- Performance Detail: 2m: 2880 St Clement's Bob Minor; 2160 Grandsire Doubles; 6 com
- Since this example is a multi-Stage Performance, the Stage Names are included in the Performance Detail. Also, although Plain Bob and Grandsire are both members of the Bob Class of Methods, Grandsire is an exception (see Section 5.F.1) in which the Bob Class Descriptor is not used. Bob therefore cannot be used in the Performance Title, but since St Clement's and Grandsire are both part of the Plain Class of Methods, Plain may optionally be included in the Performance Title (otherwise the Performance Title is 5040 Doubles and Minor).
- Performance Title: 1392 Variable Cover Spliced Cinques and Maximus
- Performance Detail: 2m: 528 Stedman Cinques; 864 Bristol Surprise Maximus; 3 com
- This example shows the format for a Performance that was Variable Cover, Spliced, and multi-Stage.

- Performance Title: 240 Triples
- Performance Detail: 1m: 240 Martyrs Link Major
- Martyrs Link has one Stationary Bell, and in this example, the Stationary Bell was not affected by any Calls. The Effective Stage of the Performance was therefore Triples. Since the Effective Stage is lower than the highest Stage of Method rung, the Performance Title and Performance Detail are constructed as above.

3. Performance Norm

An aspect of a Performance that would reasonably be assumed unless stated otherwise.

4. Ringing Style

One of the following styles in which method ringing is performed:

a) Towerbell Style: Full-circle ringing with rope and wheel, using handstroke and backstroke actions;

b) Handbell Style: Alternating upstrokes and downstrokes – usually upstrokes for handstrokes and downstrokes for backstrokes;

c) Keyboard Style: Bells sound via presses on a keyboard, such as with Online Ringing (see Section 6.A.7) or a carillon;

d) Other Style: To be specified in Performance Reports, such as ringing on the frame or tapping bells;

e) Mixed Style: More than one of the above styles were used in a Performance.

5. Distributed Ringing

Ringing in which the ringers were not all present in the same location.

6. Simulator Ringing

Ringing that involves the use of, or takes place via, a computer-based ringing simulator.

Further explanation:

The minimum involvement of a simulator is usually to play recorded or synthesised bell sounds when ringing real bells whose clappers are tied (e.g. for sound control purposes). This extends up to the simulator providing a simulation of a full ringing environment (e.g. visual and aural aspects), and possibly also simulating one or more of the ringers.

7. Online Ringing

A subset of Simulator Ringing that takes place over a computer network.

Further explanation:

Online Ringing often occurs with the ringers in different locations, in which case it is also Distributed Ringing. However, Online Ringing can also occur with all ringers in the same location, such as when a group of ringers, each with a laptop and headset, ring on a simulator such as 'Ringing Room' while all in the same room.

8. Automated Ringing

Ringing in which the sounds of one or more bells were controlled, by whatever means, without human action.

Example:

Automated Ringing occurs when a bell strikes at a time determined by a computer program.

Further explanation:

Note that fully automated ringing (i.e. with no human ringers) is not considered a reportable Performance under the Framework.

B. Content

1. A Performance Report should include the following:
   a) The Ringing Style used;
   b) Whether any Automated Ringing was used;
   c) The Location where the Performance was rung (not applicable for Distributed Ringing);
   d) In the case of Simulator Ringing / Online Ringing, or ringing facilitated by other electronic means, the name of the ringing simulator (or other means) that was used to facilitate the ringing;
   e) The date on which the Performance was rung;
   f) The Performance Title;
   g) The Performance Detail;
   h) The names of the ringers in the Performance and the number(s) of the bell(s) each rang (and in the case of Automated Ringing, make clear which bells were not rung by human ringers);
   i) In the case of Distributed Ringing, the approximate location of each ringer;
   j) The names of any umpire(s) present.

Further explanation:

In the case of Mixed Style for a), the Performance Report should include the Ringing Style used for each bell.

The Location in c) above should be sufficient to identify unambiguously where the performance was rung, eg: St Stephen, Ambridge, Borsetshire; Casterbridge Town Hall, Wessex; 23 Railway Cuttings, East Cheam.

In the case of a Performance on a boat or other moving object, the general location should be provided for c), with more detail provided in the footnote. E.g. Location: River Thames aboard the Ursula Katherine; Footnote: From Blackfriars Bridge to Woolwich.

For d), an example of other electronic means might be a distributed handbell band that is connected via telephone conference call.

If a Performance spans midnight, the date in e) above should be the date on which the Performance ended. In this case, bands may wish to include the start date (and perhaps the start or end time) in the footnote of the Performance Report.

For Distributed Ringing, the approximate location of each ringer in i) might be given as town / city, and county / state / country, as applicable.

2. A Performance Report may also include additional information such as:
   a) The society for which the Performance was rung;
   b) Details of the Composition used in the Performance, or a reference to it;
   c) The name(s) of the composer(s);

d) The time the Performance took to ring;

e) The weight and note of the tenor (towerbells) or the size and note of the tenor (handbells); and/or

f) Dedications and other footnotes.

Further explanation:

Inclusion of the Composition per b) above is encouraged in all Performance Reports. Ideally the reader of a Performance Report should have enough information to be able to reproduce the Rows that were rung.

Time per d) above is encouraged for all Lengths, and should be included for Peal Lengths.

Where applicable, tenor weight / size and note per e) above is encouraged for all Lengths, and should be included for Peal Lengths. Tenor weight / size is normally not applicable for Online Ringing and/or ringing with dummy bells.

3. The Performance Report of a Performance including one or more Methods / Variations that are not in the Methods / Variations Libraries should include the specification(s) and proposed Name(s) of those Method(s) / Variation(s).

C. Performance Norms

1. A Performance Report should state any aspect of the Performance that differed from a Performance Norm. Established Performance Norms are listed below, but the report should include any other aspect of the Performance that would not reasonably be assumed, and could be expected to alter some ringers' view of it.
2. The following are considered Norms for all reported Performances:
   a) The Performance was a Round Block that started and ended in Rounds;
   b) The Performance was a True Touch, or a Touch with Accepted Truth;
   c) The Performance was rung without interval;
   d) For Performances using Handbell Style (see Section 6.A.4), the bells were retained in hand throughout the Performance;
   e) The same person or persons rang each bell or bells continuously throughout the Performance;
   f) Neither ringers nor conductor(s) used any physical aids to memory during the Performance;
   g) No person not ringing provided any assistance in the execution of the ringing during the Performance, e.g. making calls, detecting or correcting errors;
   h) If the name of a ringing simulator (or other means) is included in the Performance Report per 6.B.1 d), the norm is that simulated sound was used; otherwise the norm is that simulated sound was not used;
   i) In any given Row of the Performance, all bells rang at the same stroke (i.e. all handstroke or all backstroke);
   j) If Cover Bell(s) were used, these were in the highest Place(s) of the Rows;
   k) Jump Changes were not used;
   l) Only one Method, Variation or Call was used to define any one Change in the Performance;
   m) A Performance with only one ringer was witnessed by an umpire;
   n) The Performance was not part of a failed longer attempt; and
   o) The Performance can be described using this Framework.

Further explanation:

For norm b), note that Accepted Truth encompasses True. See Section 3.J for definitions of these terms.

For norm c), the ethos of ringing is one of continuous performance. In the tower there is a clear distinction - the ringing either keeps going or it stops. But with handbell and keypress performances, and especially Distributed Ringing with keypresses, extended pauses are physically possible despite the ringers not intentionally stopping and re-starting. Rather than attempt to define a cutoff time, the Framework relies on the good judgement of the band in applying this norm.

For norm g), assistance such as passing someone a bottle of water, opening a window, or turning on a light is not considered assistance in the execution of the ringing.

For norm k), if Methods with Jump Changes were used, this will be evident from 6.B.1 f) or g) above, since these Methods include 'Jump' in their Titles. Therefore disclosure relating to Jump Changes is only required if Jump Calls (i.e. Calls involving Jump Changes) or Variations with Jump Changes were used, but Jump Methods (i.e. Methods involving Jump Changes) were not. (Note that Variations don't use Class Descriptors, so a Variation with Jump Changes won't have 'Jump' in its Title.)

Re: norm l), a 12-bell Performance could involve two Minor Methods rung on the front 6 and back 6 bells respectively. This is an example of a departure from norm l), and explanation should be provided in the footnotes.

For norm m), one person could (e.g.) ring a Performance of Minimus four in hand. The umpire provides corroboration that the Performance took place as reported. Note that if this norm is not followed (i.e. no umpire was present for a single person performance), this will be evident from 6.B.1 h) and j) above, so in practice no further disclosure is required.

For norm n), if, for example, an attempt to ring 7 Extents of Minor was lost in the 5th extent, the first 4 Extents are not normally considered to constitute a successful Performance. Similarly, if 7 Extents of Minor are rung but the 1st Extent is found to be false, the last 6 Extents are not normally considered to constitute a successful Performance. However, norm n) does not apply to Performances that were called round earlier than was intended at the outset of the attempt. It also does not apply to false starts, or to anything rung before or after the intended Performance.

Re: norm o), a Performance might start in the middle of a Row, or different Rows in a Performance might involve different numbers of bells ringing. These are examples of departures from norm o), and explanation should be provided in the footnotes.

7. Record Lengths

A. Requirements

1. To be included in the Central Council's register of Record Lengths, the Performance must comply with all the requirements of Sections 7.A, 7.B and 7.C.
2. All the norms listed in Section 6.C.2 applied to the Performance.
3. For a Performance using Handbell Style (see Section 6.A.4), every ringer rang at least two bells.
4. The Performance was rung by an all-human Band.

B. Notification and Verification

1. Notice of the attempt must be provided to The Ringing World to enable the notice to be published at least 14 days prior to the attempt.

Further explanation:

The Ringing World is published on Fridays, and the deadline for publication of notices is the previous Monday. For example, notice of a record length attempt occurring on Saturday 20th should be published in the RW no later than Friday 5th. This requires notice to be received by the RW on or before Monday 1st.

2. The notice must state the location, start date and start time of the attempt, the Length and the Method(s) to be rung.
3. A copy of the notice must also be provided to the Central Council at least 14 days prior to the attempt.

Further explanation:

The notice can be emailed to the Central Council at records@cccbr.org.uk.

4. Arrangement must be made for interested people to be able to listen to the Performance.
5. The ringing must be heard, and the figures of the Composition checked throughout, by competent umpire(s).

Further explanation:

The umpire(s) should do sufficient checking during the Performance to be confident that the composition was correctly called, and that there were no substantive errors or shifts. The umpire(s) should also ensure that, in their opinion, the quality of the ringing remains sufficiently high throughout the Performance for a Record Length. If the umpire(s) believe the quality has dropped below an acceptable level, and the ringing is continuing, they should let the conductor know this so the attempt can be stopped.

C. Reporting

1. A Performance Report must be sent to both The Ringing World (preferably via BellBoard) for publication, and to the Central Council.

Further explanation:

The Performance Report can be emailed to the Central Council at records@cccbr.org.uk.

2. The Performance Report must comply with the requirements of Section 6.
3. The Performance Report must include the Composition that was rung.
4. An umpire(s)' report must be sent with the Performance Report to both The Ringing World and the Central Council.

Further explanation:

The umpire(s)' report can be emailed to the Central Council at records@cccbr.org.uk.

8. Method Extension

A. Definitions

1. Extension Process

An algorithm that can be applied to the structure of a Method at one Stage to produce the structure of Methods at higher Stages. The Central Council maintains a collection of recognised Extension Processes. References in the Framework to 'Extension Process' are to those processes that are part of the Central Council's collection.

Further explanation:

The Central Council's collection of recognised Extension Processes can be found in Appendix D.

In the case of a Static Method, the structure of the Method is its place notation. In the case of a Dynamic Method, the structure of the Method is the set of rules used to generate its place notation.

2. Extension Construction

Some Extension Processes involve setting parameters, such as a mode or the subset of a Method's Changes to be copied. An Extension Construction specifies a fully qualified Extension Process – that is, it specifies an Extension Process and the parameter values to be used.

Example:

EP3-2CD/2DE

Further explanation:

An Extension Construction starts with 'EP' followed by the number of the Extension Process in the Central Council's collection of recognised Extension Processes. EP3 usually requires 4 parameters, so an EP3 Extension Construction includes these. The first number is the mode above the treble, and the letters following the number are the sections to be copied above the treble. Following the slash is the mode below the treble, followed by the sections to be copied below the treble.

The format of an Extension Construction is defined as part of the specification of the corresponding Extension Process (see Appendix D).

3. Related Methods

Two Methods at two different Stages are Related Methods if the higher Stage Method can be obtained by applying an Extension Process to the lower Stage Method.

4. Parent Method

A Method at a given Stage from which it may be possible to derive Related Methods at higher Stages using an Extension Process.

5. Extension

A Method at a higher Stage than a Parent Method that is related to the Parent Method by an Extension Process.

6. Extension Path

When two Related Methods that have the same Class Descriptor have been given the same Method Name, this establishes an Extension Path for these Methods. Additional Methods at new Stages with the same Class Descriptor that are related to these existing Methods by the same Extension Construction are on this Extension Path.

Example:

Cambridge Surprise Minor and Cambridge Surprise Major are related by Extension Construction EP3-1AB/1DE, have the same Class Descriptor ('Surprise'), and have been given the same Method Name. An Extension Path for these Methods has therefore been established.

B. Requirements

1. Related Methods with the same Class Descriptor may be given the same Method Name if the requirements of this Section 8, as well as Section 5 (Method Naming), have been met. Methods with the same Class Descriptor that are not Related may not be given the same Method Name.

Example:

Yorkshire Surprise Royal is related to Yorkshire Surprise Major by Extension Construction EP3-1AB/1FG, and they both have the same Class Descriptor ('Surprise'). These Methods may therefore be given the same Method Name (as has been done).

Further explanation:

A given Parent Method can often be extended to a number of different Extensions using different Extension Constructions. It is left to bands ringing new Extensions to decide which Extension (if any) they wish to name the same as the Parent Method.

Similarly, two or more different Parent Methods may extend, using different Extension Constructions, to the same Extension. Likewise, it is left to the band ringing the new Extension to decide which Parent Method name (if any) they wish to give to the Extension.

Note that Extension Processes often only apply to Parent Methods with a certain structure. If there is not an Extension Process that can be applied to a given Parent Method, a higher Stage Method with the same Class Descriptor may not be given the same name as the Parent Method.

2. Once an Extension Path has been established for two Related Methods with the same Class Descriptor then:
   a) New Extensions of these Methods created using the same Extension Construction, and which have the same Class Descriptor, shall be given the same Method Name; and
   b) A new Method at a different Stage with the same Class Descriptor that is not on the Extension Path shall not be given the same Name.

Example:

Yorkshire Surprise Royal is an Extension of Parent Method Yorkshire Surprise Major, they both have the same Class Descriptor ('Surprise'), and they have been given the same Name, establishing an Extension Path.

When the same Extension Construction is applied to Yorkshire Surprise Major to obtain a Maximus Method, if the resulting Maximus Method has the same Class Descriptor, it must be given the name Yorkshire because the new Method is on the Extension Path. (The resulting Maximus Method in this example is a Surprise Method, and so it has been named Yorkshire, giving Yorkshire Surprise Maximus.)

Southwark Surprise Maximus is also an Extension of Yorkshire Surprise Major by a different Extension Construction. Since an Extension Path has already been established by the naming of Yorkshire Surprise

Royal, Southwark Surprise Maximus could not have been named Yorkshire, even if the Method now known as Yorkshire Surprise Maximus had not already been named. Southwark, although related and having the same Class Descriptor, is off the Extension Path.

Further explanation:

Note that an Extension Path also applies to lower Stage Methods. For example, if Cambridge Surprise Major and Cambridge Surprise Royal established an Extension Path, since there's a Surprise Minor Method that extends using the same Extension Construction to give Cambridge Surprise Major, this Surprise Minor Method is on the Extension Path and so is also named Cambridge.

It is possible that a new Method may be on two or more existing Extension Paths. In this case, the Method Name from any of the Extension Paths may be used for the new Method.

3. There are no Extension naming restrictions per se between two Methods that have different Class Descriptors. However, when a new Extension is produced that has a different Class Descriptor from its Parent Method, the new Extension may only be given the same name as its Parent Method if this doesn't conflict with Sections 8.B.1 and 8.B.2 above as they relate to the Class Descriptor of the new Extension.

Example:

A Method designer extends Beverley Surprise Minor with an Extension Process and the resulting Extension is a Delight Royal Method. Since the Extension has a different Class Descriptor from the Parent Method, a band ringing the new Delight Royal Method may choose to name it Beverley, or give it a different name. However, the band may not name the new Method Beverley if

1. there is already a Beverley Delight Method at a different Stage to which the new Extension is not related, or
2. there is already an established Extension Path for Beverley Delight, and the new Extension is not on this Extension Path.

C. New Extension Processes

1. A band that wishes to use a different Extension Process from those included in the Central Council's collection to create a new Extension is invited to submit the proposed process to the Central Council for review.
2. The Central Council will add the new Extension Process to its collection of recognised Extension Processes if it believes it is a valid process.

D. Exceptions

1. Many higher Stage methods were named and recorded in the Methods Library before the Central Council implemented requirements for Method Extension. All such Methods retain their existing names, and the extension requirements of the Framework only apply to new Methods.
2. If a new Extension that is on an Extension Path ought, under Section 8.B, to be given the same name as its Parent Method, but that name has already been used for a different Method recorded in the Methods Library before the Central Council implemented requirements for Method Extension, then another name can be given to the new Extension.
3. If two (or more) Methods with the same Class Descriptor that are not related under the Framework have previously been given the same name, and none forms part of an Extension Path with other Method(s), then any of the Methods may be used to establish an Extension Path to other Stages.

Example:

If X Bob Major and X Bob Royal are named but not related, a Maximus Bob Method that is related to either X Bob Major or X Bob Royal could be named X, thereby establishing an Extension Path for X Bob at all other Stages.

9. Related Roles

A. Collections

1. The Central Council maintains and makes available:
    a) A Methods Library (see Section 5.A.1);
    b) A Calls Library (see Section 3.F.2);
    c) A Variations Library (see Section 5.D.2).
2. The Central Council maintains and makes available a collection of recognised Extension Processes (see Section 8.A.1).

B. Records

1. The Central Council maintains and makes available records of the Record Length rung to date in each Method / Variation, or set of Methods / Variations.

Further explanation:

'Set of Methods / Variations' refers to multi-Method Record Lengths. Record Lengths can be set for any unique combination of Methods / Variations rung in a multi-Method Performance. For example, there can be separate Record Lengths of Spliced Cambridge, Yorkshire and Superlative Surprise Major, and Spliced Cambridge, Yorkshire and Bristol Surprise Major.

Similarly, Variations can also be used to create a unique combination for a Record Length. For example, there can be separate Record Lengths for a Performance of Plain Bob and Kennington, and also for a Performance of Plain Bob and April Day.

Note that a Method is always defined at the level of its Method Title. For example, Cambridge Surprise Minor and Cambridge Surprise Major are not the same Method – they are two different Methods that are related (via a Method Extension Process). Similarly Yorkshire Surprise Major and Yorkshire Delight Major are also two different Methods – ones that happen to share the same Method Name.

2. The records separately identify:
    a) Record Lengths for the different Ringing Styles defined in Section 6.A.4;
    b) Variable Cover and non-Variable Cover Record Lengths in the same Method / Variation, or set of Methods / Variations;
    c) Records Lengths rung using simulated sound.

Further explanation:

A Variable Cover Record Length is one that included ringing a Variable Cover Composition (see Section 3.H.2).

C. Advice

1. The Central Council will endeavour to provide advice, upon request, on any aspect related to the Framework, notably Method Extension.

Further explanation:

Enquiries can be emailed to methods@cccbr.org.uk.

D. Publication

1. The Ringing World is the Central Council's official medium of publication, both in print and through its BellBoard website.
2.
    a) The Ringing World publishes Performance Reports of Quarter Peal Length and longer that it receives, categorising them according to the Lengths defined in Section 3.I;
    b) The Ringing World publishes Performance Reports of less than Quarter Peal Length that it receives when these Performances name a new Method(s);
    c) The Ringing World publishes other Performance Reports that it receives at its discretion;
    d) The Ringing World may abbreviate Performance Reports as commercial considerations necessitate.
3. The Ringing World publishes notices of Record Length attempts at least 14 days prior to the attempt, providing the notice is received in sufficient time for such publication.
4. On receiving a Performance Report of a claimed Record Length, The Ringing World waits for the Central Council's confirmation that it is a new Record Length before publishing the Report in the journal.
5. When publishing the Performance Report of a Record Length in the journal, The Ringing World includes the umpire(s)' report.

E. Analysis and Reporting

1. The Central Council produces, for each calendar year, an analysis of all Performances of Quarter Peal Length or longer that were published by The Ringing World. This analysis will either be published by The Ringing World or made available by the Central Council.
2. The exact form of the analysis is determined by those preparing it, but will likely include categorisations of Performances by:
    a) The different Lengths defined in Section 3.I;
    b) The different Ringing Styles defined in Section 6.A.4;
    c) Whether or not the ringing was Distributed and/or Online as defined in Sections 6.A.5 and 6.A.6;
    d) Whether or not the ringing was Automated as defined in Section 6.A.7;
    e) Whether or not the ringing followed the Performance Norms as defined in Section 6.C.

F. Transitional Arrangements

1. The Central Council may determine transitional arrangements relating to the implementation of the Framework, and to future updates of the Framework. For example, these arrangements may provide for retroactive recognition of Methods and Peals that were not recognised under previous Central Council Decisions or earlier versions of the Framework.